College as a Commitment Device: Parental Altruism and the Samaritan's Dilemma*

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Abstract

Why do parents invest so much in their children's college education? I propose that college serves as a commitment device for altruistic parents facing a Samaritan's dilemma. In a dynastic model where parents and adult children interact non-cooperatively, children who anticipate parental support under-save and over-consume. College mitigates this problem from the parent's perspective: by permanently raising the child's income level and lowering income risk, it shrinks the set of states where parents optimally transfer, reducing the lifetime costs of moral hazard. While the net effect on college attendance is theoretically ambiguous—anticipated transfers subsidize enrollment but also cushion non-college children against income risk—the estimated model shows that the Samaritan's dilemma distorts the composition of who attends college more than its overall level: relative to an efficient benchmark in which the parent can target transfers by education, the lack of commitment leaves too many low-return children enrolled and too few high-return children. I document a set of stylized facts from matched parent-child data (PSID and HRS) consistent with this mechanism: parents whose children attend college consume more, accumulate wealth differently, face a weaker precautionary saving motive, and are less likely to make transfers—though transfers, when they occur, are large. The structural model, estimated by the simulated method of moments, quantifies the opposing forces and the welfare cost of the commitment friction, and evaluates a free-college policy whose gains depend critically on whether parents can commit.

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1 Introduction

Parents in the United States transfer substantial resources to their adult children. Inter-vivos transfers—cash gifts, tuition payments, housing assistance—flow overwhelmingly downward, from parents to children, and persist well into adulthood (McGarry, 1999, 2016). These transfers are infrequent—in the Health and Retirement Study, only 14% of parent-child pairs report a positive transfer in any given wave—but when they occur, they are large: conditional amounts average over $8,000. Parental transfers are also a central input into children's human capital formation: families remain the primary source of college financing in the United States, and the anticipation of parental financial support is a key determinant of whether children enroll (Belley and Lochner, 2007; Brown et al., 2012; Abbott et al., 2019). From a public policy perspective, this link matters: government financial aid crowds out private parental contributions, and this crowding out is sizable precisely because family transfers are responsive to the child's enrollment decision (Abbott et al., 2019). Understanding why parents transfer, and how these transfers—both realized and anticipated—shape children's choices, is therefore fundamental both for explaining college attendance and for designing effective education policy.

This paper studies these questions through the lens of the Samaritan's dilemma (Lindbeck and Weibull, 1988). Altruistic parents who cannot credibly commit to withholding future support create an incentive problem: children who anticipate parental transfers under-save and over-consume, knowing that their parents will step in when times are bad. This strategic response forces parents into more frequent and larger transfers than they would make under commitment. The dilemma generates a feedback loop between parental altruism and children's choices—parents support their children *because* they care, but the anticipation of that support distorts the very decisions that determine how much support will ultimately be needed. Because transfers are so closely tied to the college decision, this feedback loop has direct implications for human capital investment: children who expect parental support face different incentives when choosing whether to attend college, and their college decision in turn reshapes the future pattern of transfers.

I argue that college education plays a distinctive role in this dynamic. From the parent's perspective, a dollar spent on tuition is fundamentally different from a dollar transferred directly. Tuition generates a permanent shift in the child's income level and a reduction in income risk, moving the child away from the states of the world where parental support would be triggered. A direct transfer, by contrast, provides temporary relief and may itself exacerbate moral hazard by rewarding under-saving. College thus serves as a commitment device: by investing in the child's human capital, parents can shrink the *transfer region*—the set of states where they would optimally provide support—thereby reducing the lifetime costs of the Samaritan's dilemma. The net effect on college attendance, however, is theoretically ambiguous. Anticipated transfers during the college years subsidize enrollment, raising attendance, but anticipated lifetime transfers also cushion children who forgo college, reducing their incentive to invest in human capital. The structural model quantifies these opposing forces and shows that the sign of the moral hazard effect on college varies across the ability–wealth distribution.

I formalize this mechanism in a continuous-time, non-cooperative dynastic model, building on Barczyk and Kredler (2014a,b). Parents choose consumption and a non-negative transfer flow to their adult children; children choose consumption and, at the start of the dynasty, whether to attend college. The interaction between parent and child is non-cooperative: each agent optimizes taking the other's policy as given, but the parent internalizes the child's welfare through an altruism parameter. In equilibrium, the state space is endogenously partitioned into a transfer region, where parents actively support their children, and a no-transfer region, where the child is self-sufficient. The boundary between these regions—and hence the scope of parental support—is shaped by the child's education, income risk, and accumulated wealth. College shrinks this boundary by permanently raising the child's expected earnings and lowering income risk. The child's college decision thus depends not only on the human capital return but also on how education reshapes the future financial relationship with the parent.

I document a set of stylized facts from the Panel Study of Income Dynamics (PSID) and the Health and Retirement Study (HRS) that are consistent with the model's predictions about how parent-child financial interactions differ by child education. Throughout, I compare parents whose children attended college with parents whose children did not, conditioning on parent income to isolate the role of child education from parental resources.

First, parent consumption is 8% higher when children attend college—consistent with college shrinking the transfer region and releasing resources that would otherwise be held as precautionary wealth or transferred to children. *Second*, parent wealth paths diverge by child college status: an initial reduction from tuition costs is followed by slower wealth decumulation, consistent with reduced lifetime transfer obligations. *Third*, following the framework of Boar (2021), I examine whether college attenuates dynastic precautionary saving. The point estimate implies that the negative effect of child income uncertainty on parent consumption is 62% smaller for college-educated children. Although this estimate is not statistically significant, its magnitude is economically meaningful and suggests that college reduce parents' exposure to their children's income risk. *Fourth*, within families, college reduces the probability of any transfer by 21%, but conditional on transferring, amounts are weakly larger for college children—consistent with the model's prediction that only worse shocks push college children into the transfer region. College transforms the financial relationship between parents and adult children by reducing the frequency and precautionary burden of parental support.

I develop a dynastic model of parent-child financial interactions in which parental altruism, the anticipation of support, and human capital investment are jointly determined. The model distinguishes college's role as a commitment device—reducing the moral hazard costs of the Samaritan's dilemma—from its standard human capital return, and generates predictions about how education reshapes transfers, consumption, and wealth accumulation within the family. I estimate the model by the simulated method of moments, targeting college enrollment rates by ability and wealth, the college premium, transfer patterns, and intergenerational wealth dynamics from the NLSY97 and PSID. I use the estimated model to quantify the role of the Samaritan's dilemma in shaping both college decisions and the lifetime pattern of parent-child financial interactions. Relative to an efficient benchmark in which the parent can commit to education-contingent transfers, the lack of commitment distorts the composition of college attendance—diverting resources toward marginal low-return children and away from high-return ones—more than its overall level, and it imposes a sizable welfare cost that grows with parental wealth. I show that the welfare gains from a free-college policy depend critically on whether parents can commit. These findings carry implications for the design of financial aid and transfer policies.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model. Section 4 documents the stylized facts. Section 5 describes the estimation strategy. Section 6 discusses the model fit and the role of parental altruism. Section 7 introduces the full commitment benchmark, formalizes the two opposing forces of moral hazard on college, and presents the quantitative decomposition. Section 9 concludes.

2 Related Literature

This paper is related to three branches of the literature: intergenerational transfers and family insurance, the role of parental resources in major life-cycle investments, and the determinants of college attainment.

A large body of work studies how families insure members against income risk. Since Becker and Tomes (1979, 1986), two broad motives for intergenerational transfers have been proposed: altruism, supported by evidence that transfers flow disproportionately to less well-off children (McGarry, 1999, 2016), and strategic exchange, as in Bernheim et al. (1985). The seminal tests of Altonji et al. (1992) and Hayashi et al. (1996) reject perfect insurance within the extended family, but Attanasio et al. (2018) finds substantial insurance potential between parents and adult children, and Kaplan (2012) shows that the option to move back home insures young workers against labor market risk. Most directly related to this paper, Boar (2021) identifies dynastic precautionary saving as a distinct insurance channel: parents accumulate wealth to buffer their children against permanent income uncertainty. I build on this framework and extend it by showing that college education attenuates this dynastic precautionary motive—a prediction I test empirically and embed in the structural model.

Intergenerational transfers, however, are not limited to insurance against income shocks. Parents also provide resources for large, lumpy investments such as college attendance and home purchases, and they transmit wealth through inter-vivos gifts and bequests. This broader view is important because parental transfers may affect children's outcomes not only by smoothing consumption, but also by relaxing borrowing constraints, changing portfolio choices, and shaping exposure to risk. Nishiyama (2002) studies altruistic and accidental bequests in an overlapping-generations model and shows how bequest motives affect wealth accumulation. Barczyk et al. (2023) emphasize the interaction between housing, family insurance, and the timing of transfers, showing that illiquid housing can help explain why many transfers are delayed until death. Closest to this broader mechanism, Brandsaas (2025) develops a life-cycle model with housing and parent-child transfers and finds that parental transfers account for a sizable share of young-adult homeownership by relaxing down-payment constraints, reducing housing risk, and interacting with commitment frictions. This paper builds on the insight that family transfers affect children's long-run investment, but studies this mechanism in the context of human capital accumulation.

A separate literature asks whether financial constraints bind for college attendance. Cameron and Heckman (1998) and Keane and Wolpin (2001) found that, conditional on ability, family income played a small role, but Belley and Lochner (2007) document a growing importance of family resources in the 2000s, and Lochner and Monge-Naranjo (2011) provide evidence of binding constraints for some students. Brown et al. (2012) emphasize that the expected family contribution is neither legally guaranteed nor universally provided, implying that students with similar observed family resources may face very different access to parental support. Hotz et al. (2018) provide direct evidence that parental income and housing wealth affect college attendance, parental financing, and graduation, and that parental financing may affect the subsequent indebtedness of both parents and children. On the quantitative side, Restuccia and Urrutia (2004) study how parental investments drive intergenerational earnings persistence, Caucutt and Lochner (2020) analyze the interaction between borrowing constraints and the timing of human capital investments, and Heckman and Mosso (2014) provide a comprehensive treatment of how family environments shape skill development. Closest to my setting, Abbott et al. (2019) develop a general equilibrium model with parental transfers and show that crowding out of private contributions by government financial aid is quantitatively important. My paper adds to this work by providing a mechanism—moral hazard reduction through the Samaritan's dilemma—through which parental wealth affects college enrollment beyond direct cost subsidies and paternalism.

The model builds on the quantitative literature on family dynamics without commitment, including Laitner (1988) and Lindbeck and Weibull (1988). Within this tradition, Nishiyama (2002) studies how bequests and inter-vivos transfers jointly shape the wealth distribution, and Lee and Seshadri (2019) develop a dynastic model with parental investments in children's human capital to study the intergenerational transmission of economic status. I adopt the continuous-time framework of Barczyk and Kredler (2014a,b), who show that modeling simultaneous consumption and transfer decisions in continuous time yields a well-defined equilibrium with an endogenous transfer region—the set of states where the parent optimally provides support. Barczyk and Kredler (2018) and Barczyk et al. (2023) apply this framework to long-term care and housing decisions; I extend it by embedding endogenous college decisions, which allows me to study how the transfer region—and the moral hazard it generates—interacts with education choice.

The Samaritan's dilemma that drives this moral hazard was formalized and analyzed in dynamic settings by Lindbeck and Weibull (1988) and Bruce and Waldman (1990). In my model, the distortion takes the form of strategic under-saving by the child, which raises the parent's lifetime transfer costs. College mitigates the dilemma by permanently shifting the child's income and lowering income risk, reducing the frequency with which the child enters the transfer region. The net effect on college attendance, however, is ambiguous, because anticipated transfers both subsidize enrollment and cushion the non-college path.

3 Model

This section presents a continuous-time, non-cooperative dynastic model without commitment that captures the interactions between altruistic parents and their adult children. The model extends the framework of Barczyk and Kredler (2014a,b) by incorporating endogenous college decisions, allowing me to study how family transfers affect college financial support, graduation, and parent retirement consumption. I first describe the demographic structure and income processes, then present the coupled parent-child optimization problem, the transfer variational inequality, and the college decision.

3.1 Demographics and Timing

Time is continuous. Each dynasty consists of one parent and one child who overlap for a finite period, as illustrated in Figure 1. The child enters the model at age 18, at which point the parent is 42. During the overlap period (child ages 18–42, parent ages 42–66), the parent can continuously transfer resources to the child. At age 66 ($t = T^{\text{ret}}$), the parent retires and receives social security income $\text{SS}(e_p)$ until death at age 72 ($t = T$, where $T = T^{\text{ret}} + 6$). At the parent's death, a fraction α_{beq} of remaining parent wealth transfers to the child as a bequest; the rest is absorbed by the annuity market.[1] The child then continues alone until their own retirement at 66 and death at 72.

At the beginning of the dynasty, the child makes a one-time, irreversible college decision. Children who attend college spend four years (ages 18–22) earning a fraction ℓ_C of full-time income while paying annual tuition ϕ. After college, the child enters the labor force with a permanently higher income process. Children who do not attend college enter the labor market immediately at age 18.

Both parent and child can save in a risk-free bond paying interest rate r. The child's effective rate departs from r in two cases: during college the rate is $r + \iota_s$ (a student loan premium), and outside of college the child pays $r + \iota_b$ (an unsecured borrowing premium) on any debt ($a_c < 0$). The parent faces a non-negativity constraint $a_p \geq 0$ and therefore always earns r. Markets are incomplete: no state-contingent insurance is available against the child's income risk. The parent and child are connected through the transfer flow during the overlap period and through a bequest at the parent's death: a fraction α_{beq} of remaining parent wealth transfers to the child. Inter-vivos transfers and bequests are the two channels of intergenerational resource transmission.

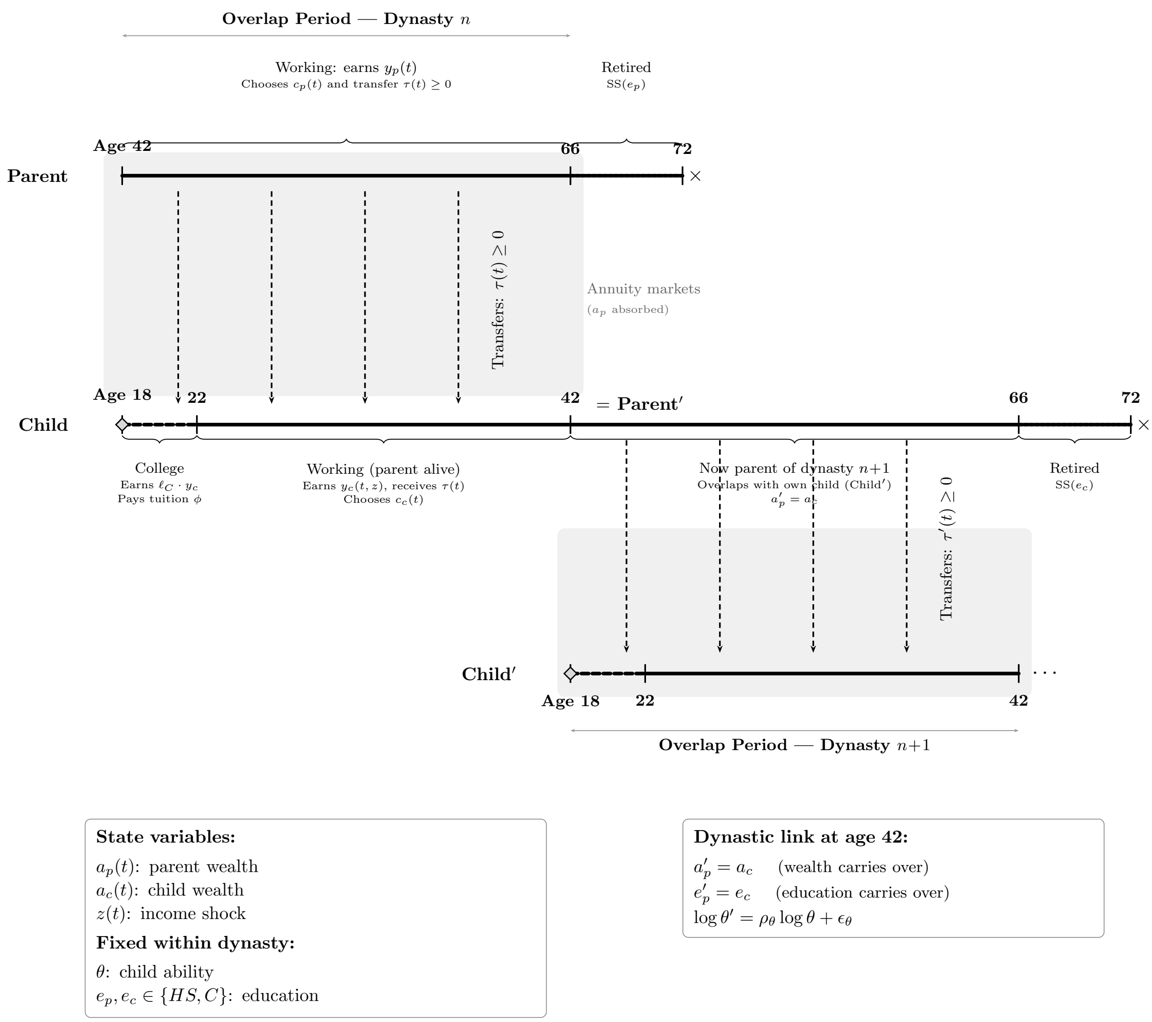
Three features of this demographic structure deserve comment. First, altruism is one-sided: the parent cares about the child's welfare, but the child does not internalize the parent's utility. Second, transfers flow only downward—from parent to child—and the overlap period is finite, ending when the parent retires at T^{ret}. The finite overlap has a natural interpretation: at age 42, the child becomes a parent in a new dynasty, redirecting resources toward their own children, while the retiring parent transitions to Social Security and annuity income. Third, the child does not support the parent in old age. A natural alternative motive for financing college is that parents invest in their children's human capital expecting financial support in return—an exchange motive in the tradition of Bernheim et al. (1985). By abstracting from

[1] The bequest fraction α_{beq} is estimated by SMM. The warm-glow motive ϕ_b gives the parent direct utility from terminal wealth, while α_{beq} governs the actual resource transfer. De Nardi (2004) estimates bequest motives implying $\alpha_{\text{beq}} \approx 0.6$–$0.8$; Boar (2021) implies ~ 0.75 in her altruism calibration.

upward transfers, the model isolates the altruism-driven Samaritan's dilemma channel from this exchange channel, allowing a clean assessment of whether moral hazard reduction alone can rationalize the observed patterns of transfers and college enrollment.

The restriction is also empirically grounded. In the United States, intergenerational transfers are overwhelmingly downward: McGarry (1999) documents that inter-vivos transfers from parents to children are an order of magnitude larger than transfers in the reverse direction, and McGarry (2016) confirms this asymmetry persists over the life cycle. Old-age consumption in the US is largely financed through Social Security, employer pensions, and private savings rather than through children's contributions—a pattern the model captures through the annuity structure and the social security income $\mathrm{SS}(e_p)$ during retirement. The exchange motive is an interesting and complementary channel, but incorporating it would require modeling two-sided altruism or strategic bargaining over old-age support, substantially complicating the equilibrium characterization without directly informing the mechanism this paper studies.

Figure 1. Dynasty Timeline



Notes: Life-cycle structure of a dynasty. The parent (top) and child (bottom) overlap for 24 years (child ages 18–42, parent ages 42–66). The shaded region indicates the overlap period during which transfers $\tau(t)$ flow from parent to child. Dashed arrows represent the transfer flow. The dashed segment on the child timeline denotes the college phase (ages 18–22). At the parent's death, a fraction α_{beq} of remaining wealth transfers to the child; the rest is absorbed by actuarially fair annuities. The lower portion illustrates the dynastic link: when the overlap period ends, the child becomes the parent of a new dynasty, carrying forward their accumulated wealth ($a_p' = a_c$), education ($e_p' = e_c$), and transmitting ability to the next generation via an AR(1) process.

3.2 State Variables and Income Processes

The state of a dynasty at any instant t during the overlap phase is $(a_p(t), a_c(t), z(t); \theta, e_p, e_c)$, where a_p and a_c are parent and child wealth, z is a stochastic productivity shock, θ is the child's cognitive ability, and $e_p, e_c \in \{HS, C\}$ denote the education levels of parent and child.

Child income. The child's labor income at time t is:

$$y_c(t) = w_{e_c} \cdot \left(\frac{\theta}{\bar{\theta}}\right)^{\lambda_{e_c}} \cdot \exp\big(\gamma(t, e_c) + z(t)\big)$$

where $h(\theta, e_c) = (\theta/\bar{\theta})^{\lambda_{e_c}}$ maps ability into human capital with an education-specific ability gradient λ_{e_c}, following Abbott et al. (2019), and $\bar{\theta} = \exp(\mu_\theta)$ is the population mean ability. The normalization by $\bar{\theta}$ ensures that the human capital component equals one at mean ability for both education groups, so that the average college premium comes entirely from the life-cycle profile and wage price ω_C, not from the level effect of $\lambda_C \neq \lambda_{HS}$ evaluated at a non-zero mean $\log \theta$. The gradient $\lambda_C > \lambda_{HS}$ implies that cognitive ability and college education are complements: higher-ability individuals benefit more from college. The education-specific wage w_{e_c} captures the relative price of education-specific labor: $w_{HS} = w$ and $w_C = w \cdot \omega_C$. In the model, the college premium is generated by three components jointly: the higher ability gradient ($\lambda_C > \lambda_{HS}$), the steeper age-income profile ($\gamma_{1,C} > \gamma_{1,HS}$), and the relative wage ω_C. The parameter ω_C is estimated via SMM to match the empirical college premium. The deterministic age-income profile $\gamma(t, e_c) = \gamma_{1,e}\, t - \gamma_{2,e}\, t^2$ is taken from Abbott et al. (2019), and $z(t)$ follows an Ornstein-Uhlenbeck process:

$$dz = -\kappa_{e_c}\, z\, dt + \sigma_{e_c}\, dW$$

with education-specific mean-reversion rate κ_{e_c} and diffusion σ_{e_c}. The mean-reversion rates are taken from Abbott et al. (2019); the diffusion coefficients σ_C and σ_{HS} are estimated via SMM to match the cross-sectional standard deviation of income by education. College transforms the child's income distribution in two ways that are central to the mechanism. First, it permanently raises the *level* of expected income through a higher ability gradient ($\lambda_C = 0.797$ vs. $\lambda_{HS} = 0.517$) and a steeper age-income profile. Second, the estimated diffusion coefficients imply substantially lower income *risk* for college graduates: the stationary standard deviation of income shocks is 0.14 for college graduates versus 0.48 for high-school graduates, so that a one-standard-deviation negative shock reduces college income by 13% but high-school income by 38%. Together, higher expected income and lower income risk keep college-educated children far from the states where parental transfers would be triggered. During college ($t \in [0, 4]$ for $e_c = C$), child income is reduced to a fraction ℓ_C of full-time earnings, reflecting part-time work while enrolled.

Parent income. Parent income is deterministic and depends on education and age: $y_p(t) = w \cdot (\theta_p/\bar{\theta})^{\lambda_{e_p}} \cdot \exp(\gamma(t, e_p))$ before retirement, and $y_p(t) = \mathrm{SS}(e_p)$ after retirement at age 66. Making parent income deterministic focuses the analysis on the child's income risk as the driver of transfers.

Ability transmission. Cognitive ability is transmitted across generations through a log-normal AR(1) process: $\log \theta' = \rho_\theta \log \theta + \mu_\theta(1 - \rho_\theta) + \epsilon_\theta$, with $\epsilon_\theta \sim N(0, \sigma_\theta^2)$ and μ_θ the unconditional mean of log ability. The drift term $\mu_\theta(1 - \rho_\theta)$ ensures that the stationary distribution of ability has mean μ_θ, generating realistic intergenerational persistence while allowing for regression to the mean.

3.3 The Coupled Parent-Child Problem

During the overlap period, parent and child simultaneously choose consumption and savings, while the parent additionally chooses a non-negative transfer flow $\tau(t) \geq 0$. The interaction is non-cooperative: each agent optimizes taking the other's policy as given, but the parent internalizes the child's welfare through the altruism parameter $\eta > 0$.

The continuous-time formulation follows Barczyk and Kredler (2014a), who show that simultaneity of moves in continuous time yields a well-defined equilibrium—an advantage over discrete-time formulations where the timing protocol (who moves first within a period) can affect the equilibrium outcome.

3.3.1 Parent's Problem

The parent chooses consumption c_p and transfer flow $\tau \geq 0$ to maximize lifetime utility, weighting the child's instantaneous utility by η. The parent's value function $V^p(a_p, a_c, z; t)$ satisfies:

$$\rho V^p = \max_{c_p,\, \tau \geq 0} \Big\{ u(c_p) + \eta\, u(c_c^*) + V_{a_p}^p [r a_p + y_p - c_p - \tau] \\ + V_{a_c}^p [\tilde{r}_c\, a_c + y_c - c_c^* + \tau] + V_z^p(-\kappa z) + \tfrac{1}{2}\sigma^2 V_{zz}^p + V_t^p \Big\} \tag{1}$$

where ρ is the continuous-time discount rate, $u(c) = c^{1-\gamma}/(1-\gamma)$ is CRRA utility with $\gamma = 1.5$, and c_c^* is the child's equilibrium consumption policy, taken as given by the parent. Consumption is bounded below by a floor $\underline{c} = \$3{,}900$ (in 2016 dollars), representing the effective consumption guaranteed by means-tested government transfers (SSI, SNAP, Medicaid), following De Nardi (2004). The terms involving V_z^p and V_{zz}^p capture the effect of income uncertainty on the parent's value function.

3.3.2 Child's Problem

The child chooses consumption c_c to maximize own lifetime utility:

$$\rho V^c = \max_{c_c} \Big\{ u(c_c) + V_{a_c}^c [\tilde{r}_c\, a_c + y_c - c_c + \tau^*] \\ + V_{a_p}^c \big[r\, a_p + y_p - c_p^* - \tau^* \big] + V_z^c(-\kappa z) + \tfrac{1}{2}\sigma^2 V_{zz}^c + V_t^c \Big\} \tag{2}$$

where τ^* and c_p^* are the parent's equilibrium policies, taken as given. The child tracks parent wealth a_p because it affects expected future transfers: wealthier parents are more likely to provide support.

3.3.3 Transfer Decision

The parent's optimal transfer is characterized by a complementarity condition:

$$\tau^* \geq 0, \quad V_{a_p}^p \geq V_{a_c}^p, \quad \tau^* \Big(V_{a_p}^p - V_{a_c}^p \Big) = 0 \tag{3}$$

This partitions the state space into two regions. In the *no-transfer region* ($\mathcal{N}$), the parent's marginal value of own wealth exceeds the marginal value of child wealth ($V^p_{a_p} > V^p_{a_c}$), and $\tau^* = 0$. In the *transfer region* ($\mathcal{T}$), these marginal values are equalized ($V^p_{a_p} = V^p_{a_c}$) and $\tau^* > 0$. The boundary between $\mathcal{N}$ and $\mathcal{T}$ is a free boundary, endogenously determined as part of the equilibrium. The parent transfers when the child is relatively poor (low a_c), faces negative productivity shocks (low z), or when the parent is relatively wealthy (high a_p).

This variational inequality is the continuous-time analog of the transfer decision in discrete-time dynastic models, but with two key advantages. First, transfers are a continuous flow rather than lump-sum payments, eliminating the timing discreteness that affects transfer amounts in discrete models. Second, the free boundary provides a clean characterization of the states where moral hazard is operative.

3.3.4 Consumption Policies and Wealth Dynamics

The first-order conditions for consumption are:

$$u'(c_p^*) = V^p_{a_p}, \tag{4}$$

$$u'(c_c^*) = V^c_{a_c}. \tag{5}$$

Under CRRA utility, these invert to $c_p^* = (V^p_{a_p})^{-1/\gamma}$ and $c_c^* = (V^c_{a_c})^{-1/\gamma}$. Wealth evolves as:

$$\dot{a}_p = r\, a_p + y_p(t) - c_p - \tau, \tag{6}$$

$$\dot{a}_c = \tilde{r}_c\, a_c + y_c(t, z) - c_c + \tau, \tag{7}$$

where $a_p \geq 0$ and the child's effective interest rate is

$$\tilde{r}_c = \begin{cases} r + \iota_s & \text{during college } (t \in [0,4],\ e_c = C), \\ r + \iota_b\, \mathbf{1}\{a_c < 0\} & \text{otherwise,} \end{cases}$$

so the child earns r on savings and pays $r + \iota_b$ on any unsecured debt. During college the borrowing constraint is relaxed to $a_c \geq -\bar{b}$ (see Section 3.7).

3.4 Child-Alone Problem and the Dynastic Link

When the overlap period ends (parent age 66, child age 42), the child solves a consumption-savings problem with their own accumulated wealth for the remainder of their working life:

$$\rho\, V^{c,\text{alone}} = \max_{c_c} \left\{ u(c_c) + V^{c,\text{alone}}_{a_c}[\tilde{r}_c\, a_c + y_c(t,z) - c_c] + V^{c,\text{alone}}_{z}(-\kappa z) + \tfrac{1}{2}\sigma^2 V^{c,\text{alone}}_{zz} + V^{c,\text{alone}}_{t} \right\} \tag{8}$$

At this point, the child becomes a parent in a new dynasty: their own child is born with ability θ' drawn from the intergenerational transmission process (Section 3), and the new parent-child pair solves the coupled problem described above. The child's accumulated wealth a_c at the end of the overlap period becomes the initial parent wealth a_p for the new dynasty. The solution of (8) at $t = 0$ provides the terminal condition for the coupled system.

3.5 Terminal Conditions

At the end of the overlap period ($t = T^{\text{ret}}$, parent age 66, child age 42), the coupled system terminates. The parent continues into retirement until death at $t = T$ (age 72), receiving social security income $\text{SS}(e_p)$ and decumulating wealth through annuity purchases. However, because no transfers flow after T^{ret}, the parent and child solve independent problems from this point onward. The boundary conditions at $t = T^{\text{ret}}$ are:

$$V^p(a_p, a_c, z; T^{\text{ret}}) = \phi_b \frac{a_p^{1-\gamma}}{1-\gamma} + \eta \cdot V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p,\ z;\ 0), \tag{9}$$

$$V^c(a_p, a_c, z; T^{\text{ret}}) = V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p,\ z;\ 0). \tag{10}$$

Because the parent and child are no longer interacting after $t = T^{\text{ret}}$, each agent's continuation is independent. The child's value function transitions to the child-alone problem (8), evaluated at $t = 0$ of the child-alone clock—the moment the overlap ends. Equation (10) simply states this continuity.

The parent's terminal value (9) has two components. The first, $\phi_b\, u(a_p)$ with $\phi_b \geq 0$, is a warm-glow term following De Nardi (2004) that captures the parent's direct utility from holding wealth at the end of the overlap period. Although the parent continues to live and consume during retirement ($t \in [T^{\text{ret}}, T]$), the coupled problem is solved only over the overlap horizon; the warm-glow term summarizes the parent's continuation value from their own wealth during retirement. The second component, $\eta \cdot V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p, z; 0)$, reflects the parent's altruistic concern for the child's continuation utility after the overlap ends. The child's continuation value is evaluated at effective wealth $a_c + \alpha_{\text{beq}}\, a_p$, where $\alpha_{\text{beq}} \in [0, 1]$ is the fraction of parent terminal wealth that transfers to the child as a bequest at death. This bequest channel creates two motives for parental wealth accumulation: the warm-glow motive ϕ_b and the altruistic motive through α_{beq}, which disciplines the intergenerational wealth correlation. The parent also has a forward-looking motive for inter-vivos transfers during the overlap: by transferring resources while alive, the parent raises a_c at $t = T^{\text{ret}}$ and thereby improves the child's post-overlap continuation value.

3.6 College Decision

At the beginning of the dynasty ($t = 0$), the child decides whether to attend college. I model this as a logit choice. Let V^c_C, V^c_{HS} denote the child's equilibrium continuation values and V^p_C, V^p_{HS} the parent's continuation values under college and high school, evaluated at the initial state $(a_p, a_c = 0, z = 0)$. The psychic cost of college follows the Abbott et al. (2019) linear-in-log specification with an idiosyncratic shock:

$$\kappa(\theta, \varepsilon_\kappa) = [\kappa_0 + \kappa_\theta \log \theta + \sigma_\kappa\, \varepsilon_\kappa]^+ \cdot \bar{V},$$

where $\varepsilon_\kappa \sim N(0, 1)$ is drawn once per individual at $t = 0$, $[\cdot]^+$ denotes $\max(\cdot, 0)$, and $\bar{V} = \text{mean}_{a_p,\theta} |V^c_C - V^c_{HS}|$ is a normalization constant that makes the dimensionless parameters κ_0, κ_θ, and σ_κ comparable in scale to the college value gap. The gradient $\kappa_\theta < 0$ implies that higher-ability individuals face lower psychic costs (following Cunha et al. 2005 and Heckman et al. 2006), and the shock $\sigma_\kappa \varepsilon_\kappa$ generates college attendance heterogeneity among low-ability types. Define the utility indices:

$$U_C(\varepsilon_\kappa) \equiv V^c_C - \kappa(\theta, \varepsilon_\kappa) + \psi\, V^p_C, \qquad U_{HS} \equiv V^c_{HS} + \psi\, V^p_{HS},$$

where $\psi \geq 0$ is a paternalism weight that captures the child's internalization of parental welfare when choosing education, following Abbott et al. (2019).[2] The population college probability at each state (a_p, θ) integrates over the psychic cost shock:

$$\Pr(e_c = C \mid a_p, \theta) = \int \frac{\exp\left[U_C(\varepsilon)/\sigma_{cd}\right]}{\exp\left[U_C(\varepsilon)/\sigma_{cd}\right] + \exp\left[U_{HS}/\sigma_{cd}\right]} \phi(\varepsilon)\, d\varepsilon, \tag{11}$$

where σ_{cd} is the scale parameter of the Type-I extreme value taste shock and the integral is computed via Gauss-Hermite quadrature with seven nodes.

The college decision is endogenous in two senses. First, parental altruism affects the child's continuation value: parents who anticipate supporting their child during and after college raise the value of attending. Second, the continuation values under $e_c = C$ and $e_c = HS$ incorporate the entire future trajectory of transfers, which depends on the child's income path and hence on education. This creates a channel through which parent wealth affects college enrollment beyond the direct cost channel. Importantly, anticipated transfers affect both V_C and V_{HS}: transfers subsidize college enrollment (raising V_C) but also cushion non-college children against income risk (raising V_{HS}). The net effect of the Samaritan's dilemma on the college decision is therefore theoretically ambiguous; Section 7.1 formalizes this result.

3.7 College Financial System

I model the college financial system following Abbott et al. (2019). The annual cost of attending college is $\phi = \$6{,}700$, reflecting average net tuition (after institutional discounts) from the NLSY97. During college, students work a fraction $\ell_C = 0.56$ of full-time hours.

Financial aid is modeled as a continuous function of parental wealth through an expected family contribution (EFC) schedule. The annual grant received by a student with parent wealth a_p is:

$$g(a_p) = \bar{g} \cdot \max\left(0,\ 1 - \frac{a_p}{\bar{a}_{\text{efc}}}\right)$$

where $\bar{g}$ is the maximum grant (for families with zero wealth) and $\bar{a}_{\text{efc}}$ is the wealth level at

[2] The paternalism weight ψ enters only the college decision, not the child's consumption-savings problem during the overlap period. It captures the idea that children partially internalize the cost their education choice imposes on parents—for example, through family pressure or social norms—without altering the non-cooperative structure of the within-period transfer game.

which aid phases out completely. Both $\bar{g}$ and $\bar{a}_{\text{efc}}$ are estimated by SMM. This continuous specification captures the essential feature of the federal financial aid system: grants decline smoothly with family resources, generating a wealth-dependent effective price of college.

During college, students can borrow up to a cumulative limit $\bar{b}(a_p) = \bar{b}_0 + b_{\text{slope}} \cdot a_p$ at a uniform blended student loan rate $r + \iota_s$, following Abbott et al. (2019). The base limit $\bar{b}_0 = \$23{,}000$ reflects the average federal student loan cap; the collateral channel $b_{\text{slope}} \geq 0$ allows children of wealthier parents to borrow more (e.g., through parental co-signing), and is estimated by SMM. This is implemented by allowing $a_c \geq -\bar{b}(a_p)$ during $t \in [0, 4]$, with the constraint $a_c \geq 0$ restored after graduation. The child's wealth dynamics during college become:

$$\dot{a}_c = (r + \iota_s)\, a_c + \ell_C \cdot y_c(t, z) - c_c - \phi_{\text{net}}(a_p) + \tau, \qquad a_c \geq -\bar{b}(a_p)$$

where $\phi_{\text{net}}(a_p) = \phi - g(a_p)$ is net tuition. This system generates heterogeneity in the effective cost of college across the wealth distribution. Parental transfers supplement institutional aid, and as Abbott et al. (2019) document, government grants can crowd out private contributions.

3.8 Equilibrium

A Markov-Perfect Equilibrium (MPE) consists of value functions V^p, V^c, consumption policies c_p^*, c_c^*, a transfer policy τ^*, and a college enrollment rule e_c^*, for each (θ, e_p, e_c) pair, such that: (i) given c_c^* and τ^*, the parent's HJB (1) and the variational inequality (3) are satisfied; (ii) given c_p^* and τ^*, the child's HJB (2) is satisfied; (iii) after the overlap ends ($t > T^{\text{ret}}$), $V^{c,\text{alone}}$ satisfies (8); (iv) the college rule follows from (11); and (v) boundary conditions (9)–(10) and borrowing constraints hold at all times.

The equilibrium is solved backward: first the child-alone problem, then the coupled system during the overlap period, and finally the college choice at $t = 0$. Within the coupled system, the transfer region and consumption policies are determined jointly through policy iteration on the variational inequality. Appendix I derives the equilibrium properties, including the Samaritan's dilemma distortion in continuous time, and Appendix J describes the numerical solution algorithm.

3.9 Model Predictions: How College Affects the Transfer Region

The model generates predictions about how college alters the financial relationship between parents and adult children. The central mechanism operates through the transfer region $\mathcal{T}$—the set of states where the parent optimally transfers resources to the child. College education shifts the boundary of this region, and the consequences propagate to consumption, wealth, and precautionary behavior.

Why college shrinks the transfer region. Recall that the parent transfers when the marginal value of own wealth equals the marginal value of child wealth ($V^p_{a_p} = V^p_{a_c}$). Because the parent weighs the child's utility by η in the objective (1), $V^p_{a_c}$ is increasing in η: more altruistic parents have a larger transfer region. This condition is more likely to bind when the child is relatively poor or faces negative income shocks. College raises the child's expected income through two channels: a higher ability gradient ($\lambda_C > \lambda_{HS}$) and a steeper life-cycle earnings profile. The resulting increase in child wealth pushes states away from the transfer boundary, so college-educated children spend less time in $\mathcal{T}$. In addition, the estimated diffusion coefficients imply substantially lower income risk for college graduates ($\sigma_C \ll \sigma_{HS}$), so negative shocks are smaller in magnitude and less likely to push the child into the transfer region. Together, higher expected income and lower income risk shrink $\mathcal{T}$ for college-educated children relative to high-school-educated children.

Figure 2 illustrates this mechanism in the (a_c, a_p) state space. The boundary $\partial\mathcal{T}$ separates states where transfers are positive (to the left) from states where the parent saves independently (to the right). College shifts this boundary to the left: at any given level of parent wealth, the child needs to be poorer before transfers become optimal. The region $\mathcal{T}_{HS} \setminus \mathcal{T}_C$—states where a high-school child would receive transfers but a college child would not—represents the scope for moral hazard that college eliminates.

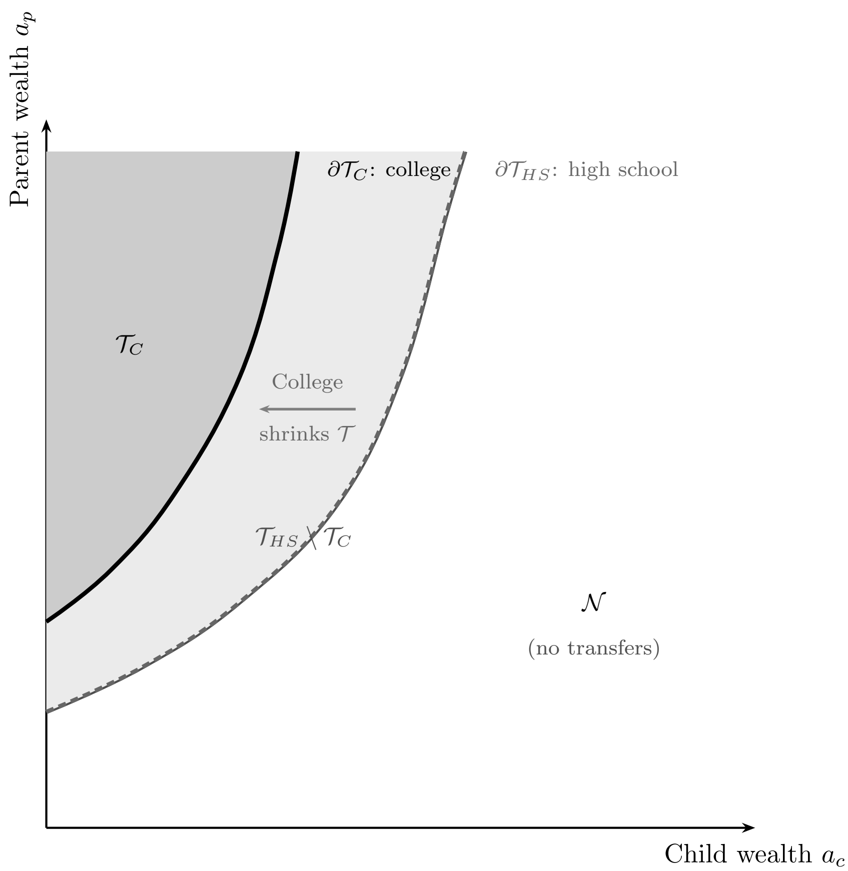


Figure 2. Transfer Region Boundary: College vs. High School

Notes: Schematic illustration of the transfer region $\mathcal{T}$ in the (a_c, a_p) state space. In the transfer region (shaded), the parent optimally transfers to the child ($\tau > 0$). The solid boundary corresponds to a college-educated child; the dashed boundary corresponds to a high-school-educated child. College shrinks $\mathcal{T}$ because higher expected income keeps child wealth further from the transfer boundary.

Four testable predictions. The contraction of the transfer region generates specific predictions about observable outcomes, all conditional on parent income:

1. *Parent consumption is higher when the child attends college.* A smaller transfer region means that parents spend less of their lifetime making transfers. The resources that would have flowed to the child under the high-school path are instead available for own consumption.

2. *Parent wealth paths diverge by child education.* In the short run, parents of college children pay tuition costs, reducing their wealth. In the long run, however, the smaller

transfer region means fewer ongoing transfers, and parent wealth recovers and eventually exceeds the high-school path.

3. *College attenuates the precautionary saving motive.* Parents hold precautionary wealth against the contingency of supporting their children after negative income shocks. Because college raises the child's income level and lowers income risk, parents of college children face a smaller precautionary burden. This prediction links directly to the option value of the transfer region: even if transfers are rarely observed, the *possibility* of having to transfer affects parent saving behavior.

4. *College reduces the probability of entering the transfer region, but transfers conditional on entry may be larger.* Because college shifts the boundary of $\mathcal{T}$ to the left—requiring a larger negative shock to trigger transfers—the probability of entering $\mathcal{T}$ falls. However, conditional on entry, the states reached are further from the boundary (the child must be in a worse position for the transfer condition to bind), so the optimal transfer τ^* may be weakly larger. The model therefore predicts a negative extensive-margin effect and a zero or slightly positive intensive-margin effect. Combined with Prediction 3, this implies that the consumption and wealth gains from college far exceed the reduction in realized transfer flows—the gap reflects the option-value channel operating through precautionary behavior rather than through observed transfers.

I now turn to the data to test these predictions.

4 Empirical Evidence

This section documents a set of stylized facts from matched parent-child data (PSID, HRS, and NLSY97) that are consistent with the model's predictions. Each stylized fact compares outcomes for parents whose children attended college with those whose children did not, conditioning on parent income to isolate the role of the child's education from the parent's own resources. I begin with the relationship between child college status and parent consumption, then examine wealth accumulation, the dynastic precautionary saving channel, and finally the structure of inter-vivos transfers.

4.1 Data

The empirical analysis draws on three data sources. The Panel Study of Income Dynamics (PSID) provides matched parent-child panels with consumption, income, wealth, and inter-vivos transfer data from 1999 onward. I link parents to adult children using the Family Identification Mapping System (FIMS) and restrict the sample to parents aged 50+ and children aged 26+, yielding 8,944 observations across 2,338 parent-child pairs. The Health and Retirement Study (HRS) offers a larger sample of older parent-child pairs (19,179 pairs, 98,861 observations) with detailed transfer information, which I use to study the extensive and intensive margins of inter-vivos transfers. The National Longitudinal Survey of Youth 1997 (NLSY97) provides cognitive ability scores (ASVAB) and parental wealth for 5,400 individuals, which I use to construct college graduation rates by ability and wealth quartile for the SMM estimation (Section 5). Throughout, the key explanatory variable is child college attainment—an indicator for holding a bachelor's degree or higher—and I condition on within-cohort parent income quartiles to isolate the role of child education from parental resources. All monetary variables are in 2016 dollars, winsorized at the 1st and 99th percentiles. Appendix A documents the sample construction pipeline, and Appendix A.5 provides detailed variable definitions.

4.1.1 Summary Statistics

Tables 1–2 report summary statistics for the PSID sample, stratified by average household income group. Table 1 presents parent demographics and key financial variables: parents in higher income groups are older on average, more likely to hold a college degree, more likely to own their home, and hold substantially more wealth. Total wealth ranges from $103,000 for the lowest income group to $966,000 for the highest, with home equity accounting for roughly half of wealth for low-income families but a smaller share for wealthy families. Total consumption expenditures increase monotonically with income, from $36,000 to $73,000. A detailed breakdown of the wealth portfolio is provided in Appendix Table E.1, and a consumption decomposition by expenditure category in Appendix Table F.1. Table 2 presents child-level statistics, including the college attendance rate among 18–24 year olds by parental income group, which rises steeply with parental resources.

Table 1. Parent Demographics and Financial Characteristics by Income Group (PSID)

	By Average HH Income Group				
	$30–60	$60–100	$100–160	$160+	All
Panel A: Demographics					
Age of head	45.6	45.2	46.8	49.1	46.3
	(16.2)	(14.4)	(13.4)	(13.2)	(14.6)
Children in household	1.3	1.2	1.1	1.0	1.1
	(1.4)	(1.3)	(1.2)	(1.1)	(1.3)
Total children	2.1	2.0	2.1	2.2	2.1
	(1.3)	(1.0)	(1.0)	(1.0)	(1.1)
Head has college degree	0.07	0.19	0.42	0.69	0.29
	(0.26)	(0.39)	(0.49)	(0.46)	(0.45)
White	0.44	0.63	0.77	0.85	0.64
	(0.50)	(0.48)	(0.42)	(0.36)	(0.48)
Black	0.45	0.29	0.18	0.09	0.28
	(0.50)	(0.45)	(0.38)	(0.28)	(0.45)
Homeowner	0.50	0.71	0.84	0.85	0.70
	(0.50)	(0.45)	(0.36)	(0.36)	(0.46)
Panel B: Income, Wealth, and Consumption					
Household income	44,223	76,829	120,169	242,277	102,569
	(21,535)	(37,964)	(51,447)	(253,591)	(121,904)
Head labor income	23,523	40,976	64,728	123,504	53,930
	(19,369)	(28,162)	(42,902)	(123,970)	(64,202)
Total wealth	102,555	200,522	367,076	965,941	326,288
	(230,260)	(377,059)	(536,534)	(1,052,497)	(612,283)
Total consumption	36,426	46,131	57,243	73,130	50,076
	(18,374)	(20,389)	(24,489)	(33,277)	(26,252)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. Income groups defined by parent average real household income (2016 dollars) observed between ages 25 and 60. Panel A reports demographics; Panel B reports income, wealth, and consumption. “Children in household” is the number of children under age 18 in the parent’s PSID family unit; “Total children” is the total number of children linked to the parent household via the Family Identification Mapping System (FIMS). All monetary variables in 2016 dollars, winsorized at the 1st and 99th percentiles. A detailed wealth portfolio breakdown is in Appendix Table E.1.

Table 2. Children Descriptive Statistics by Parent Income Group (PSID)

	By Parent's Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
Panel A: Child Demographics					
Age	17.9	15.9	16.7	17.4	16.9
	(19.0)	(15.7)	(13.6)	(13.4)	(16.0)
Number of siblings in sample	1.9	1.6	1.5	1.6	1.6
	(1.7)	(1.2)	(1.2)	(1.1)	(1.4)
College-age (18–24)	0.14	0.14	0.16	0.17	0.15
	(0.34)	(0.35)	(0.36)	(0.38)	(0.36)
Panel B: Education					
Years of education (age 18+)	13.7	13.7	13.8	14.4	13.9
	(11.8)	(9.9)	(7.8)	(8.1)	(9.7)
Max years of education (age 25+)	16.1	15.7	15.9	16.5	16.0
	(14.8)	(11.5)	(9.9)	(8.9)	(11.9)
Currently in college	0.05	0.06	0.08	0.10	0.07
	(0.21)	(0.23)	(0.27)	(0.29)	(0.25)
Ever attains post-secondary ed.	0.33	0.37	0.47	0.56	0.41
	(0.47)	(0.48)	(0.50)	(0.50)	(0.49)
Panel C: College Rate Among 18–24 Year Olds					
College attendance rate	0.32	0.40	0.47	0.54	0.42
	(0.47)	(0.49)	(0.50)	(0.50)	(0.49)
College-age observations	4308	4984	4270	2907	16 469
Total child–year observations	31 731	35 484	27 026	16 962	111 203
Unique children	5680	5560	3784	2186	17 210

Notes: Means with standard deviations in parentheses. Each observation is a child-year. Children linked to parents via FIMS. Income groups refer to parent average real household income (2016 dollars). "Years of education" reported conditional on age ≥ 18; "Max years of education" conditional on age ≥ 25. "College attendance rate" computed conditional on the child being aged 18–24. "Ever attains post-secondary ed." indicates maximum observed years of education exceeds 12. Years of education directly observed only from 2013 onward; for earlier waves, college attendance inferred from an age-education proxy.

4.2 Parent Consumption and Child College Status

The model predicts that, conditional on their own income and wealth, parents whose children attend college should consume more (Prediction 1 in Section 3.9). College shifts children out of the transfer region $\mathcal{T}$ by permanently raising their income, freeing parents from contingent obligations and raising their consumption even when no transfers are actually flowing.

I test this prediction using the following regression on the PSID sample at the parent-year level:

$$C_{p,t} = \beta_0 + \beta_1 \text{ShareCollege}_{p,t} + \beta_2 N_{p,t} + \delta_{Q_p^{inc}} + \beta_X \mathbf{X_{p,t}} + \varepsilon_t + \epsilon_{p,t}$$

where $C_{p,t}$ is household consumption (in 2016 dollars) of parent p at time t, $\text{ShareCollege}_{p,t}$ is the fraction of parent p's children with a bachelor's degree or higher, $N_{p,t}$ is the total number of children, $\delta_{Q_p^{inc}}$ are parent income quartile fixed effects (within-cohort ranking), and $\mathbf{X_{p,t}}$ is a comprehensive set of controls: parent total wealth, non-financial wealth, household income, wealth quartile, labor force participation, household size, head's birth year, education, U.S. state, a cubic in head's age, housing tenure, and race. The specification includes year fixed effects ε_t.

Table 3 presents the results. Column 1 shows that a higher share of college-educated children is associated with significantly higher parent consumption. Columns 2 and 3 use alternative definitions—any child with a BA, or all children with a BA—yielding consistent results. Column 4 uses consumption levels rather than logs; the magnitude implies economically meaningful differences. These effects are net of parent income, wealth, and education, suggesting they are not driven by parental resources alone.

Table 3. Parent Consumption and Child College Status (PSID)

	(1) Log C	(2) Log C	(3) Log C	(4) Level C
Share of Children with BA+	0.081***			2470.032***
	(0.019)			(866.461)
Number of Children	0.014*	0.010	0.016*	169.725
	(0.008)	(0.008)	(0.008)	(287.036)
Any Child with BA+		0.080***		
		(0.017)		
All Children with BA+			0.056***	
			(0.017)	
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Parent income Q FE	Yes	Yes	Yes	Yes
Observations	15,830	15,830	15,830	15,831
R^2	0.558	0.558	0.557	0.357

Notes: OLS regressions of parent household consumption on child college indicators. "Share of Children with BA+" is the fraction of children with a bachelor's degree or higher. Controls include a cubic in parent age, household income, non-housing wealth, total wealth, wealth quartile, and fixed effects for year, education, parent income quartile, gender, housing tenure, state, labor force status, race, household size, and birth year. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 4 estimates the effect separately by parent income quartile, confirming that the positive relationship between child college status and parent consumption holds across the income distribution.

Table 4. Parent Consumption and Child College Status, by Parent Income Quartile (PSID)

	Parent Income Quartile			
	(1)	(2)	(3)	(4)
	Q1 (Poorest)	Q2	Q3	Q4 (Richest)
Share of Children with BA+	0.136***	0.053*	0.016	0.041
	(0.040)	(0.030)	(0.025)	(0.029)
Number of Children	0.010	0.003	0.018*	0.001
	(0.013)	(0.010)	(0.009)	(0.013)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	4,342	3,830	3,931	3,727
R^2	0.442	0.326	0.355	0.415

Notes: OLS regressions of log parent household consumption on share of children with BA+, estimated separately by parent income quartile. Controls as in Table 3 excluding parent income quartile dummies. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

4.3 Parent Wealth Accumulation

The consumption premium documented above has a natural counterpart in wealth accumulation. The model predicts that parent wealth paths should diverge by child education: a short-run decline from tuition costs, followed by long-run recovery as the smaller transfer region reduces ongoing obligations (Prediction 2 in Section 3.9). I test this prediction by examining how parent wealth differs by child college status, conditioning on parent income.

Using PSID data, I estimate:

$$W_{p,t} = \beta_0 + \beta_1 \text{ShareCollege}_{p,t} + \beta_2 N_{p,t} + \delta_{Q_p^{inc}} + \beta_X \mathbf{X_{p,t}} + \varepsilon_t + \epsilon_{p,t}$$

where $W_{p,t}$ is parent p's total real wealth at time t and other variables are defined as in Section 4.2. The model makes a nuanced prediction about wealth. Three forces operate simultaneously on parent wealth when a child does not attend college: (i) a *precautionary* *accumulation* motive—parents save more because the child may need future transfers; (ii) a *Samaritan's dilemma* distortion—inside $\mathcal{T}$, where $V^p_{a_p} = V^p_{a_c}$, each additional dollar of parent wealth partially leaks to the child through transfers, reducing the effective return on saving; and (iii) actual *transfer outflows* that deplete accumulated wealth. Forces (i) and (ii) affect desired wealth accumulation in opposite directions, while force (iii) reduces realized wealth. Which force dominates is a quantitative question that depends on the altruism parameter η and the size of $\mathcal{T}$—precisely the objects the structural model is designed to identify.

Table 5 presents the cross-sectional results. Columns 1 and 2 show that the association between college and parent wealth is positive but statistically imprecise, consistent with the opposing forces partially offsetting. The log wealth specification (column 3) yields a significant positive association. Column 4 adds an interaction between college share and parent age (centered at 65) to test whether the relationship changes over the lifecycle.

Table 5. Parent Wealth and Child College Status (PSID)

	(1) Total Wealth	(2) Non-Housing Fin.	(3) Log Wealth	(4) Wealth (Age Int.)
Share of Children with BA+	43601.955	12219.029	0.328***	73804.841
	(64206.109)	(60376.529)	(0.056)	(78817.996)
Number of Children	33086.211	32690.416	0.024	32683.531
	(31431.443)	(30135.932)	(0.020)	(31367.974)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Parent income Q FE	Yes	Yes	Yes	Yes
Observations	16,068	16,068	14,644	16,068
R^2	0.225	0.187	0.615	0.227

Notes: OLS regressions of parent real wealth on the share of children with a bachelor's degree or higher. Controls include a cubic in parent age, household income, and fixed effects for year, education, parent income quartile, gender, housing tenure, state, labor force status, race, household size, and birth year. "Age Int." adds interactions between college share and a quadratic in parent age (centered at 65). Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 6. Parent Wealth and Child College Status: Fixed Effects (PSID)

	(1) Total Wealth (FE)	(2) Non-Housing Fin. (FE)
Share of Children with BA+	153524**	147930**
	(60734)	(60562)
Number of Children	74799**	72682**
	(33045)	(32939)
Parent FE	Yes	Yes
Year FE	Yes	Yes
Observations	16,068	16,068
Within R^2	0.007	0.006

Notes: Parent fixed-effects regressions of real wealth on the share of children with a bachelor's degree or higher and the number of children. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 6 presents parent fixed-effects estimates exploiting within-parent variation over time: the share of college-educated children is associated with \$153,524 higher total wealth (significant at the 5% level). Rather than confirming a simple directional prediction, this result reflects the net outcome of the three competing forces. The structural model decomposes this: by simulating counterfactuals in which individual forces are shut down, I quantify how much of the observed wealth differential is driven by precautionary accumulation, how much by the Samaritan's dilemma distortion, and how much by realized transfer flows (Section 7).

4.4 Dynastic Precautionary Saving

The previous two stylized facts document large differences in parent consumption and wealth associated with child college status. The model identifies the mechanism: college raises the child's expected income and lowers income risk ($\sigma_C \ll \sigma_{HS}$), jointly shrinking the transfer region. This prediction is closely related to the dynastic precautionary savings channel identified by Boar (2021), who shows that parents accumulate precautionary savings to insure their children against permanent income risk—a 10% increase in a child's income uncertainty reduces parent consumption by 0.76%. Following her framework, I test whether college attenuates this dynastic precautionary motive.

There is strong independent evidence that college graduates face lower earnings risk. Meghir and Pistaferri (2004) estimate that the variance of earnings growth is 0.065 for college graduates compared to 0.103 for high school graduates—a 37% reduction.

I test this prediction using the following specification, which extends equation (9) of Boar (2021) with a child education interaction:

$$\ln c_{it}^{p} = \beta_0 + \beta_t + \beta_1 \sigma_{hs}^{p} + \beta_2 \sigma_{hs}^{c} + \beta_3 \sigma_{hs}^{c} \times \mathbf{1}\{e_c = \text{College}\} + \mathbf{X}_{it}^{p} \beta_4 + \mathbf{X}_{it}^{c} \beta_5 + \varepsilon_{it} \tag{12}$$

where σ_{hs}^{c} is the child's permanent income uncertainty, constructed following Boar's methodology by estimating sector-specific (age $\times$ industry-occupation) forecast errors of lifetime earnings, and σ_{hs}^{p} is the parent's own uncertainty. The key coefficient is β_3: the model predicts $\beta_2 < 0$ (higher child uncertainty reduces parent consumption) and $\beta_3 > 0$ (college attenuates this effect).

Table 7 presents the results. Column 1 replicates Boar's baseline specification in the PSID sample, confirming that higher child income uncertainty reduces parent consumption ($\hat{\beta}_2 < 0$). Column 2 adds the education interaction: the positive $\hat{\beta}_3 = 0.012$ indicates that college attenuates the dynastic precautionary motive, though the estimate is imprecise ($p = 0.11$). The net effect of child uncertainty for college-educated children is $\hat{\beta}_2 + \hat{\beta}_3 = -0.020 + 0.012 = -0.008$, substantially smaller than the effect for non-college children ($\hat{\beta}_2 = -0.020$), representing a 62% reduction. College substantially weakens parents' exposure to child income risk—consistent with the model's prediction that college shrinks the transfer region (Prediction 3 in Section 3.9).

Table 7. Dynastic Precautionary Savings and Child Education

	(1) Baseline	(2) Education Interaction
σ^p_{hs} (Parent uncertainty)	-0.0149***	-0.0149***
	(0.00564)	(0.00565)
σ^c_{hs} (Child uncertainty)	-0.0116***	-0.0198***
	(0.00370)	(0.00696)
$\sigma^c_{hs} \times \mathbf{1}\{\text{College}\}$		0.0123
		(0.00770)
Parent controls	Yes	Yes
Child controls	Yes	Yes
Year FE	Yes	Yes
Parent income Q FE	No	Yes
Observations	12897	12897
R^2	0.782	0.782

Notes: Dependent variable is log OECD-equivalized non-durable parent consumption. σ^p_{hs} and σ^c_{hs} are permanent income uncertainty measures constructed as $\ln\sqrt{\text{Var}(\varepsilon)+1}$, where ε is the level-scale forecast error from sector-specific age-income profiles estimated on the full PSID income sample, following Boar (2021). Column 2 adds parent income quartile fixed effects and the college interaction. Additional controls: year fixed effects, parent and child age dummies, parent education, marital status, race, and family size. Standard errors clustered by parent household $\times$ year in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Across Sections 4.2–4.4, college is associated with 8% higher parent consumption (roughly $2,500 per year), divergent wealth paths, and a substantially weaker precautionary saving response to child income risk. The dynastic precautionary saving result identifies the mechanism directly: college reduces the sensitivity of parent consumption to child income uncertainty by 62%, from -0.020 to -0.008. This operates through the *option value* of the transfer region: even when transfers are not currently flowing, the *possibility* of future transfers depresses parent consumption. College shrinks this region, releasing resources.

4.5 Inter-Vivos Transfers

A natural question arises: if college affects parent consumption and wealth so strongly, do we observe correspondingly large differences in actual transfers? As Prediction 4 in Section 3.9 anticipates, the answer requires decomposing transfers into their extensive and intensive margins—and this decomposition maps directly onto the model's transfer-region structure.

The key feature of inter-vivos transfers in the HRS is that most parent–child pairs report zero transfers in any given wave: only 14.4% of observations involve a positive transfer from parent to child (Table 8). Conditional on a positive transfer, however, amounts are substantial—\$8,161 on average, with a median of \$2,897 and a 90th percentile of \$18,394. This pattern—a large mass at zero with a heavy right tail—is precisely what the model predicts. The transfer region $\mathcal{T}$ is entered infrequently, but when entered, transfers are non-trivial insurance events.

Table 8. Distribution of Inter-Vivos Transfers from Parents to Children (HRS)

Pr(Transfer > 0)	0.144
E(Transfer | Transfer > 0)	\$8,161
Median(Transfer | Transfer > 0)	\$2,897
P75(Transfer | Transfer > 0)	\$7,968
P90(Transfer | Transfer > 0)	\$18,394
E(Transfer, unconditional)	\$1,172
Observations	548,239
Observations with Transfer > 0	78,541

Notes: Wave-level statistics for parent-to-child transfers from HRS 1992–2014 (Family-Kids file, waves 1–12). Sample restricted to parent age ≥ 50, child age ≥ 26. Transfer amounts in real 2016 US dollars. Unconditional mean includes zeros.

I decompose the within-family college effect along these two margins. Table 9 reports linear probability models for Pr(Transfer > 0) with parent fixed effects:

$$\mathbf{1}[IVT_{ijt} > 0] = \beta_0 + \beta_1 \text{College}_j + \gamma_t + \beta_X \mathbf{X_{jt}} + \delta_{Q_i^p} + \varepsilon_i + \epsilon_{ijt}$$

College reduces the probability of any parent-to-child transfer by 3.1 percentage points—a 21% reduction relative to the mean transfer probability of 14.4% (Table 9, Column 1). Column 2 interacts college with parent income quartile, revealing a steep gradient: the effect of college on transfer probability is -1.0 pp for Q1 parents, -2.8 pp for Q2 ($-0.010 - 0.018$), -3.4 pp for Q3, and -4.2 pp for Q4. Wealthier parents have larger transfer regions ex ante—they can afford to support children in more states of the world—so college shrinks their transfer region by more. Meanwhile, college *increases* the probability that the child transfers back to the parent by 1.3 pp (Column 3), consistent with college raising the child's permanent income.

Table 9. Extensive Margin: Effect of College on Transfer Probability (HRS)

	Extensive Margin (Linear Probability)		
	(1)	(2)	(3)
	Pr(Parent $\rightarrow$ Child)	Pr(Parent $\rightarrow$ Child)	Pr(Child $\rightarrow$ Parent)
College (BA+)	-0.031***	-0.010***	0.013***
	(0.002)	(0.003)	(0.001)
College $\times$ Q2		-0.018***	
		(0.004)	
College $\times$ Q3		-0.024***	
		(0.004)	
College $\times$ Q4		-0.032***	
		(0.005)	
Parent FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes
Dep. Var. Mean	0.144	0.144	0.024
Observations	548,164	548,164	521,676
Within R^2	0.012	0.012	0.002

Notes: Linear probability models. Dependent variable is an indicator for any positive transfer in the HRS wave. All specifications include parent fixed effects, wave dummies, child cohort controls, and parent income quartile dummies. Standard errors clustered at parent level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 10 conditions on positive transfers and confirms the model's second prediction for this margin. Conditional on being in the transfer region, college children receive \$557 more per transfer event (Column 1, $p < 0.10$). Column 2 reveals heterogeneity: for Q1 parents, college children receive \$906 *less* conditional on entry, but for wealthier parents the effect is positive and large—\$958 more for Q2 ($-906 + 1{,}864$), and \$826 more for Q4. This pattern is consistent with the boundary-shift mechanism: because college moves the transfer boundary $\partial\mathcal{T}$ to the left in the (a_c, a_p) space, the states that push a college-educated child into $\mathcal{T}$ involve worse realizations of child income. Conditional on entry, the child is further from the boundary, and the optimal transfer is therefore larger—particularly for wealthier parents whose transfer regions are larger and whose boundary shift from college is more pronounced. Column 3 shows that college children also transfer \$390 more back to parents conditional on any reverse transfer, consistent with their higher permanent income.

Table 10. Intensive Margin: Transfer Amount Conditional on Positive Transfer (HRS)

	Intensive Margin (Amount | Transfer > 0)		
	(1)	(2)	(3)
	Parent → Child	Parent → Child	Child → Parent
College (BA+)	557*	-906*	390***
	(288)	(486)	(119)
College × Q2		1864***	
		(607)	
College × Q3		965*	
		(542)	
College × Q4		1732**	
		(684)	
Parent FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes
Dep. Var. Mean	8161	8161	3554
Observations	78,541	78,541	12,383
Within R^2	0.002	0.002	0.009

Notes: OLS regressions on the subsample with positive transfers. Dependent variable is the real transfer amount (2016 US\$). All specifications include parent fixed effects, wave dummies, child cohort controls, and parent income quartile dummies. Standard errors clustered at parent level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

The reduction in transfer probability implied by college—3.1 percentage points—saves roughly \$253 per year in expected transfers ($0.031 \times \$8{,}161$), an order of magnitude smaller than the \$2,500 consumption gain documented in Section 4.2. The gap between observed transfer savings and the consumption effect is consistent with the option-value channel identified in Section 4.4: college reduces the precautionary burden parents face, releasing resources beyond what flows through actual transfers.

5 Estimation

I estimate the model in three stages. In the first stage, I set parameters that are well-identified from external data or the existing literature. In the second stage, I estimate the income process independently and map discrete-time estimates to their continuous-time counterparts. In the third stage, I estimate the remaining seventeen structural parameters—altruism, psychic costs, the warm-glow bequest motive, the bequest fraction, the discount rate, income diffusion, the college wage scale, paternalism, transitory income shock variances, the financial aid schedule, and the collateral channel—by the simulated method of moments (SMM), matching 35 data moments.

5.1 Externally Set Parameters

Table 11 reports the parameters set in the first stage. The coefficient of relative risk aversion is $\gamma = 1.5$, following Abbott et al. (2019). The annual real interest rate is $r = 0.03$, following Daruich and Kozlowski (2020). The continuous-time discount rate ρ is estimated internally by SMM to match the wealth-to-income ratio and wealth distribution moments. The college costs and borrowing limits are set as described in Section 3.7; the financial aid schedule parameters ($\bar{g}$, $\bar{a}_{\text{efc}}$) are estimated by SMM (see below). The intergenerational ability parameters (ρ_θ, σ_θ, μ_θ) are taken from Abbott et al. (2019). Retirement income is estimated directly from PSID data on households where the respondent is retired and over age 67, computing the average sum of Social Security, SSI, disability, and employer pension income by education group.

Table 11. Externally Set Parameters

Parameter	Description	Value	Source
Preferences			
r	Real interest rate	0.03	Daruich and Kozlowski (2020)
γ	CRRA risk aversion	1.5	Abbott et al. (2019)
$\underline{c}$	Consumption floor ($000s/yr)	3.9	De Nardi (2004)
Demographics (years)			
T_{overlap}	Parent–child overlap (18–42)	24	—
$T_{\text{retire},p}$	Parent retirement (66–72)	6	—
T_{alone}	Child alone (42–66)	24	—
$T_{\text{retire},c}$	Child retirement (66–72)	6	—
Ability gradients			
λ_C	Returns to ability, college	0.797	Abbott et al. (2019)
λ_{HS}	Returns to ability, HS	0.517	Abbott et al. (2019)
Earnings process (annual)			
ρ_C^{ann}	Persistence, college	0.966	Abbott et al. (2019)
σ_C^{ann}	Innovation s.d., college	0.08	Abbott et al. (2019)
ρ_{HS}^{ann}	Persistence, HS	0.952	Abbott et al. (2019)
σ_{HS}^{ann}	Innovation s.d., HS	0.07	Abbott et al. (2019)
$\bar{w}$	Avg. annual earnings	$70k	Census
College costs			
ϕ_C	Annual tuition (before aid)	$6.7k	Abbott et al. (2019)
τ_C	Time worked in college	0.56	Census
Intergenerational ability			
ρ_θ	Persistence of log ability	0.36	Abbott et al. (2019)
σ_θ	Std. dev. of ability innovation	0.40	Abbott et al. (2019)
μ_θ	Mean of log ability	1.30	Abbott et al. (2019)
Financial aid & borrowing			
$\bar{b}_0$	Student loan limit (base)	$23.0k	Federal
ι_s	Student loan premium	0.031	Abbott et al. (2019)
ι_b	Unsecured borrowing premium	0.064	Abbott et al. (2019)
Retirement income (annual)			
SS_C	Soc. Security + pension, college	$14.3k	PSID
SS_{HS}	Soc. Security + pension, HS	$13.0k	PSID

Notes: Dollar amounts in 2016 dollars. Persistence mapped to the continuous-time mean-reversion rate via $\kappa_e = -\ln \rho_e^{\text{ann}}$; the OU diffusion coefficients σ_C and σ_{HS} are estimated by SMM (Table 13). Income process persistence from Abbott et al. (2019), Table 3.2 (males). The financial aid schedule $g(a_p) = \bar{g} \cdot \max(0, 1 - a_p/\bar{a}_{\text{efc}})$ has parameters $\bar{g}$ and $\bar{a}_{\text{efc}}$ estimated by SMM (Table 13). Retirement income from PSID (respondent retired, age > 67). Student loan base limit is cumulative over four college years; the effective limit is $\bar{b}(a_p) = \bar{b}_0 + b_{\text{slope}} \cdot a_p$, where b_{slope} is estimated by SMM (Table 13). Interest rate premia: effective student rate is $r + \iota_s = 0.061$; effective unsecured borrowing rate is $r + \iota_b = 0.094$.

5.2 Income Process Estimation

The income process in continuous time is an Ornstein-Uhlenbeck process $dz = -\kappa_e z\, dt + \sigma_e\, dW$ with education-specific parameters. I take the deterministic age profile and the income-shock persistence directly from the male estimates in Abbott et al. (2019), mapping the persistence to its continuous-time mean-reversion counterpart κ_e. The education-specific diffusion coefficients σ_C and σ_{HS}, which govern income risk, are *not* taken from Abbott et al. (2019): they are estimated jointly with the other structural parameters by the simulated method of moments, disciplined by the education-specific income standard deviations.[3]

In discrete time, log labor income is decomposed as in Abbott et al. (2019): $\log \epsilon_j = \lambda_e \log \theta + \gamma_{e,j} + z_j$, where λ_e is the education-specific ability gradient. The ability gradient and age-income profile are taken from Abbott et al. (2019), who estimate λ_e from the NLSY79 by regressing log hourly wages on log AFQT scores, separately by education group, and estimate the deterministic age-income profile $\gamma_{e,j}$ from PSID data using a quadratic in age. The income shock persistence is also taken from Abbott et al. (2019), who estimate it by minimum distance on the autocovariance structure of wage residuals by age and education:

$$z^e_{iat} = \log y_{it} - \widehat{f^e(a_{it})} - \hat{\lambda}_e \log(\text{AFQT}_i)$$

$$z^e_{iat} = \rho^{\text{ann}}_e \, z^e_{i,a-1,t-1} + \eta^e_{iat}$$

$$\eta^e_{iat} \sim N(0, (\sigma^e_\eta)^2), \;\; z^e_{i0t} \sim N(0, (\sigma^e_{z_0})^2)$$

I map the discrete-time persistence to its continuous-time mean-reversion counterpart κ_e by matching the autocovariance structure; Appendix D derives the mapping formula. Table 12 reports the discrete-time estimates. The resulting mean-reversion rates are similar across education groups ($\kappa_{HS} = 0.049$, $\kappa_C = 0.035$). The education-specific diffusion coefficients σ_C and σ_{HS}, by contrast, are not mapped from Abbott et al. (2019) but are estimated by the simulated method of moments together with the other structural parameters, disciplined by the education-specific income standard deviations. The estimates imply substantially lower income risk for college graduates ($\sigma_C \ll \sigma_{HS}$), consistent with the evidence in Meghir and Pistaferri (2004).

[3] Abbott et al. (2019) estimate the income process separately by gender. I use their male estimates throughout, as the model abstracts from gender differences.

Table 12. Income Process and Age Profile

Ability Gradient (Abbott et al., 2019, Table 3.1, Males)		
	High-School	College Graduate
λ_e	0.517	0.797
Age Profile (Abbott et al., 2019, Table E.1, Males)		
	High-School	College Graduate
$\gamma_{1,e}$	0.0673	0.1197
$\gamma_{2,e} \times 1000$	-0.670	-1.191
Income Process (Abbott et al., 2019, Table 3.2, Males)		
	High-School	College Graduate
ρ_z^{ann}	0.952	0.966
σ_η^{ann}	0.07	0.08
σ_{z_0}	0.14	0.16

Notes: All discrete-time parameters are taken directly from Abbott et al. (2019). Top panel: ability gradient from their Table 3.1. Middle panel: age-income profile from their Table E.1. Bottom panel: income process parameters (ρ_e^{ann}, σ_η^e, $\sigma_{z_0}^e$). The continuous-time mean-reversion rates κ_e are obtained from ρ_e^{ann} via the mapping described in the text; the OU diffusion coefficients σ_C and σ_{HS} are estimated by SMM (Table 13).

Figure D.1 in Appendix D illustrates the implied life-cycle income profiles by education, including the deterministic age-income component, expected income for a median-ability worker, the role of OU shock volatility, and the complementarity between ability and education.

5.3 Ability Gradient

The ability gradient λ_e in the human capital function $h(\theta, e) = (\theta/\bar{\theta})^{\lambda_e}$ governs the returns to cognitive ability by education level. I take the estimates of λ_e directly from Abbott et al. (2019), who estimate them from the NLSY79 by regressing log hourly wages (purged of the PSID age profile) on log AFQT scores, separately by education group. The key estimates are $\hat{\lambda}_{HS} = 0.517$ and $\hat{\lambda}_C = 0.797$, implying that a one-log-point increase in ability raises college earnings by 80% but high-school earnings by only 52%. This complementarity between ability and education is central to the mechanism: the college premium is strongly increasing in ability, so removing parental transfers has heterogeneous effects across the ability distribution. These parameters are reported in Table 11.

5.4 SMM Estimation

The remaining seventeen parameters—parent altruism η, psychic cost parameters κ_0, κ_θ, and σ_κ, the college choice scale parameter σ_{cd}, the discount rate ρ, the warm-glow weight ϕ_b, the bequest fraction α_{beq}, the education-specific income diffusion parameters σ_C and σ_{HS}, the college wage scale ω_C, the paternalism weight ψ, the transitory income shock standard deviations $\sigma_{\varepsilon,C}$ and $\sigma_{\varepsilon,HS}$, the financial aid parameters $\bar{g}$ and $\bar{a}_{\text{efc}}$, and the collateral channel slope b_{slope}—are estimated by the simulated method of moments (SMM; McFadden, 1989; Pakes and Pollard, 1989). SMM is well suited to this setting because the model does not admit closed-form expressions for the moments of interest: the coupled HJB system, the endogenous transfer region, and the logit college choice must be solved numerically. Following Duffie and Singleton (1993) and Lee and Ingram (1991), I simulate the model at each candidate parameter vector and match the simulated moments to their data counterparts.

Let $\mathbf{x} \in \mathbb{R}^{17}$ denote the parameter vector. For each $\mathbf{x}$, I solve the model and simulate $M = 4{,}000$ dynasties, computing 35 moments $\mathbf{m}^{\text{sim}}(\mathbf{x})$. The estimator minimizes a block-diagonal percentage-deviation loss:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \sum_{g=1}^{G} \frac{w_g}{n_g} \sum_{m \in g} \left[\frac{m_m^{\text{sim}}(\mathbf{x}) - m_m^{\text{data}}}{m_m^{\text{data}}} \right]^2, \tag{13}$$

where the 35 moments are partitioned into $G = 5$ groups: college attendance rates ($n_g = 16$, $w_g = 3$), income process ($n_g = 3$, $w_g = 1$), transfers ($n_g = 5$, $w_g = 1$), intergenerational wealth ($n_g = 10$, $w_g = 1$), and decumulation ($n_g = 1$, $w_g = 1$). The group-size normalization $1/n_g$ ensures that larger groups do not mechanically dominate. College moments receive a weight of three because they are the core target of the model; downweighting them leads to well-fitting wealth distributions but implausibly flat college-ability gradients. While an optimal weighting matrix based on the inverse of the variance-covariance of the moments would be asymptotically efficient, it requires a first-stage consistent estimate that is computationally expensive in this setting. The block-diagonal weighting provides a transparent alternative and is standard in the structural life-cycle literature (Abbott et al., 2019; Boar, 2021).

5.4.1 Target Moments

The 35 target moments are organized into five groups, each chosen to provide leverage on specific parameters.

College graduation rates (16 moments). College graduation rates by ability quartile $\times$ parent wealth quartile, from the NLSY97. These moments form the core of the estimation: they trace out how college attainment varies across the joint distribution of ability and parental resources, which is precisely the variation the model is designed to explain.

Income process moments (3 moments). The high-school-to-college mean income ratio and income standard deviations by education, computed from individual gross earnings in the NLSY97.[4]

Transfer moments (5 moments). The mean transfer amount and the college attendance probability by parent wealth quartile. The transfer-probability moments use college attendance as a proxy for the parent-to-child transfer gradient, constructed from the PSID intergenerational sample.[5]

Aggregate wealth (1 moment). The wealth-to-income ratio, which disciplines the overall level of wealth accumulation in the economy.

Intergenerational wealth distribution (10 moments). From the PSID linked sample: the parent–child wealth rank–rank slope ($\hat{\beta}^{rr} = 0.372$), mean child wealth conditional on parent wealth quartile, median wealth for parents and children, the variance of the inverse

[4]Because each agent in the model represents a single individual—not a household—the income targets use individual gross earnings (wages and salary) from the NLSY97, not household income. This is consistent with the approach in Boar (2021) and Lee and Seshadri (2019), who also target individual labor income in single-agent dynastic models. The income ratio (0.69) is computed from the cross-section of individuals aged 25–37 to maximize sample size, while the standard deviations ($\sigma_{HS} = 27.7$K and $\sigma_C = 41.7$K) are computed from ages 30–37 to match the model's age-35 income snapshot. All figures are in thousands of 2016 dollars, deflated by the CPI-U. The sample conditions on positive earnings above $5,000 and defines college graduates as those with a bachelor's degree or higher.

[5]I link children to their parents using the PSID Family Identification Mapping System (FIMS) and observe children as household heads or spouses at ages 35–40, with parent wealth measured when the child was 17–18. All wealth is in thousands of 2016 dollars (CPI-adjusted). Appendix A details the sample construction.

hyperbolic sine of wealth for each generation, and the parent wealth decumulation rate (ages 60–70).[6]

5.4.2 Identification

The model is overidentified: seventeen parameters are estimated from 35 moments. While all moments jointly contribute to the estimation, the identification logic rests on distinct sources of variation for each parameter group.

Altruism (η). The altruism parameter is identified primarily from the wealth gradient in college attendance, holding ability constant. A higher η increases the size of the transfer region $\mathcal{T}$, raising the lifetime transfer burden parents face when children do not attend college. This makes college more attractive to wealthy parents, steepening the wealth gradient. The transfer probability moments by parent wealth quartile provide additional leverage: higher altruism implies more frequent transfers, with a wealth gradient shaped by the transfer region boundary.

Psychic cost (κ_0, κ_θ, σ_κ). The psychic cost parameters are identified from the ability gradient in college attendance, holding wealth constant. The constant κ_0 governs the overall college rate, the log-ability gradient κ_θ determines how steeply college attendance rises with ability, and the shock scale σ_κ controls the dispersion of college choices among low-ability types: a larger σ_κ generates more college attendance at the bottom of the ability distribution by giving some individuals favorable psychic cost draws.

Discount rate (ρ). The discount rate is pinned down by the aggregate wealth-to-income ratio and median wealth levels. A lower ρ (more patient agents) generates more wealth accumulation, raising both statistics.

Warm-glow weight (ϕ_b). The warm-glow motive is identified from the parent wealth decumulation rate: a stronger warm-glow motive (ϕ_b) slows wealth drawdown in retirement,

[6] The rank–rank slope of 0.372 is consistent with the range in the literature: Charles and Hurst (2003) estimate an intergenerational wealth elasticity of 0.37, and Pfeffer and Killewald (2018) find rank correlations of 0.40 or higher when children are measured at ages 45–64. My estimate falls in the middle because children in my sample are observed at ages 35–40.

as parents derive utility from holding wealth at T. The decumulation rate of $+2.5\%$ per year (parents ages 60–70 are still accumulating) is consistent with evidence that pre-retirees continue saving due to precautionary motives and warm-glow utility (De Nardi, 2004; Hurd, 1990). The intergenerational rank-rank slope and conditional child wealth moments provide additional leverage on ϕ_b by disciplining how parental wealth transmits across generations.

Income diffusion (σ_C, σ_{HS}). The education-specific diffusion parameters are identified from the income standard deviation moments and their interaction with the transfer and college moments: σ_C and σ_{HS} govern how frequently income shocks push children into the transfer region, affecting both transfer flows and the option value of college.

College choice scale (σ_{cd}). The scale parameter governs the dispersion of college choices around the deterministic value function difference. It is identified from the overall fit of the 16 college graduation rates: a larger σ_{cd} flattens the relationship between the value function gap $U_C - U_{HS}$ and the probability of attendance.

College wage scale (ω_C). The relative price of college labor ω_C is identified primarily from the college-to-high-school income ratio: without this scaling, the structural premium from ability gradients ($\lambda_C > \lambda_{HS}$) and the steeper college life-cycle profile would generate a college premium far exceeding the data.

Paternalism (ψ). The paternalism weight is identified from the interaction between parent wealth and college attendance. A positive ψ amplifies the wealth gradient by giving the child an additional reason to attend college: internalizing the parent's preference for reduced transfer costs.

Transitory shocks ($\sigma_{\varepsilon,C}$, $\sigma_{\varepsilon,HS}$). The transitory income shock variances add cross-sectional income dispersion without affecting the persistent OU component. They are identified from the cross-sectional income standard deviations, complementing the diffusion parameters σ_C and σ_{HS}.

Financial aid ($\bar{g}$, $\bar{a}_{\mathbf{efc}}$). The maximum grant $\bar{g}$ and the EFC wealth phase-out threshold $\bar{a}_{\text{efc}}$ are identified from the wealth gradient in college attendance: they govern how steeply the effective price of college rises with parent wealth.

Bequest fraction ($\alpha_{\mathbf{beq}}$). The bequest fraction is identified from the intergenerational wealth moments: the rank–rank slope, conditional child wealth by parent wealth quartile, and the variance of child wealth. A higher α_{beq} strengthens the intergenerational wealth correlation by channeling more parent wealth to the child at death.

Collateral channel ($b_{\mathbf{slope}}$). The collateral channel slope is identified from the wealth gradient in college attendance, particularly at the top of the parent wealth distribution. A positive b_{slope} relaxes borrowing constraints for children of wealthy parents, raising college enrollment among high-wealth families.

Table 13. SMM-Estimated Parameters

Parameter	**Description**	**Value**
Preferences		
β	Discount factor (annual)	0.87
σ_{cd}	Type-I EV scale (college decision)	1.73
Parental altruism & paternalism		
η	Altruism weight	0.48
ϕ_b	Warm-glow terminal wealth weight	0.45
α_{beq}	Bequest fraction (share of parent wealth to child)	0.70
ξ	Paternalism weight in college decision	0.02
College psychic cost (AGMV-style)		
κ_0	Psychic-cost constant	4.1
κ_θ	Psychic-cost gradient in $\log(\theta)$	-3.0
σ_κ	Psychic-cost idiosyncratic shock S.D.	1.20
Income process		
ω_C	Relative price of college labor	0.48
σ_C	College income diffusion (OU)	0.038
σ_{HS}	High-school income diffusion (OU)	0.151
$\sigma_{\varepsilon,C}$	Transitory income shock S.D., college	0.204
$\sigma_{\varepsilon,HS}$	Transitory income shock S.D., high school	0.070
College financial aid (continuous EFC)		
$\bar{g}$	Maximum annual grant ($000s)	2.0
$\bar{a}_{\text{efc}}$	EFC phase-out wealth threshold ($000s)	236.4
b_{slope}	Collateral channel slope ($\bar{b} = \bar{b}_0 + b_{\text{slope}} \cdot a_p$)	0.00

Notes: All 17 parameters are estimated jointly by the simulated method of moments (SMM). σ_{cd} is the scale of i.i.d. type-I extreme-value shocks in the college attendance choice. ξ is the paternalism weight: the parent's value difference $V_C^p - V_{HS}^p$ enters the child's college decision. $\kappa(\theta, \varepsilon_\kappa) = \kappa_0 + \kappa_\theta \log(\theta) + \sigma_\kappa \varepsilon_\kappa$ is the psychic cost of college (AGMV specification), decreasing in ability. ϕ_b governs the warm-glow utility $\phi_b\, u(a_p)$ from terminal wealth. α_{beq} is the fraction of parent terminal wealth that transfers to the child as a bequest at death; the remainder is absorbed by the annuity market (see Section 3). The same altruism weight η enters both the parent's flow utility and the terminal condition. ω_C is the relative price of college labor; σ_C and σ_{HS} are the diffusions of the education-specific OU income processes, and $\sigma_{\varepsilon,C}$, $\sigma_{\varepsilon,HS}$ the standard deviations of i.i.d. transitory income shocks. Net college tuition is $\phi - \bar{g} \max\{0,\, 1 - a_p/\bar{a}_{\text{efc}}\}$: $\bar{g}$ is the maximum grant and $\bar{a}_{\text{efc}}$ the parental-wealth level at which grants fully phase out. b_{slope} governs the wealth-dependent collateral channel in the student borrowing limit: $\bar{b}(a_p) = \bar{b}_0 + b_{\text{slope}} \cdot a_p$. Parameters set outside the estimation are reported in the companion table of externally calibrated parameters.

Notes: Parameters estimated by simulated method of moments. The objective function minimizes the percentage-deviation distance between 35 simulated moments and their data counterparts. See text for the identification argument.

6 Results

6.1 Model Fit

Table 14 reports the model's fit to the 35 targeted moments. Panel A shows college graduation rates by ability and parent wealth quartile. The model captures the primary features of the data: college graduation rates increase with both ability and parent wealth, and the wealth gradient is steepest for low-ability children. The model slightly underpredicts college attainment for the highest-ability children, who in the data attend college at near-universal rates that leave little room for variation. Panel B reports income moments and Panel C transfer moments. Panel D reports the intergenerational wealth distribution moments constructed from the PSID parent–child linked sample that discipline the warm-glow weight ϕ_b and the intergenerational wealth transmission channel.

The model captures the broad patterns of the data: college graduation rates increase strongly with ability within each wealth quartile, and the overall graduation rate is close to the data average. However, the model underpredicts the wealth gradient in college attainment. In the data, low-ability children with parents in the top wealth quartile graduate at a rate of 33%, compared to 19% for those in the bottom quartile—a gap of 14 percentage points. The model generates only a modest wealth gradient for this group (22% vs. 26%), suggesting that the current calibration underestimates the role of direct cost subsidies or that additional channels (e.g., neighborhood effects, information frictions) contribute to the empirical wealth gradient beyond the altruism mechanism. The model performs better along the ability dimension: for Q1 parents, the model generates a gap of 22 percentage points between the lowest and highest ability quartiles (22% to 44%), compared to 34 percentage points in the data (19% to 53%).

Table 14. Targeted Moments: Data and Model

Panel A: College Attainment by Parent Wealth and Child Ability (NLSY97)				
Wealth Quartile \ Ability Quartile	1	2	3	4
1	0.24	0.37	0.43	0.46
	(0.19)	(0.24)	(0.33)	(0.53)
2	0.21	0.40	0.47	0.49
	(0.24)	(0.30)	(0.42)	(0.53)
3	0.21	0.37	0.45	0.46
	(0.26)	(0.39)	(0.51)	(0.63)
4	0.27	0.43	0.43	0.50
	(0.33)	(0.46)	(0.62)	(0.74)

Panel B: Income Moments		
	Model	Data
High-School/College mean income ratio	0.62	0.69
High-School income S.D. ($k)	27.7	27.7
College income S.D. ($k)	40.5	41.7

Panel C: Transfer Moments		
	Model	Data
Mean transfer ($k/yr)	8.2	6.2
Pr(transfer > 0) — Wealth Q1	0.24	0.22
Pr(transfer > 0) — Wealth Q2	0.26	0.31
Pr(transfer > 0) — Wealth Q3	0.33	0.50
Pr(transfer > 0) — Wealth Q4	0.47	0.73

Panel D: Intergenerational Wealth Moments		
	Model	Data
Wealth–income ratio	1.75	1.87
Rank–rank slope (IG wealth)	0.18	0.37
$E[a_c \mid \text{parent wQ1}]$ ($k)	74.4	67.1
$E[a_c \mid \text{parent wQ2}]$ ($k)	98.6	112.1
$E[a_c \mid \text{parent wQ3}]$ ($k)	98.1	177.6
$E[a_c \mid \text{parent wQ4}]$ ($k)	130.0	340.4
Median wealth, parents ($k)	60.7	85.4
Median wealth, children ($k)	62.4	51.2
Var(IHS wealth), parents	4.68	7.25
Var(IHS wealth), children	4.57	13.48
Parent wealth growth rate	0.033	0.025

Notes: Comparison of 35 targeted moments in the data and the estimated model. Panel A reports college graduation rates by parent wealth quartile (rows) and child ability quartile (columns); numbers without parentheses are model moments, numbers in parentheses are data moments (NLSY97). Panel B: income moments. Panel C: transfer moments; Pr(transfer > 0) uses college attendance by parent wealth quartile as proxy for the extensive margin of financial transfers, from the PSID parent–child linked sample. Panel D: intergenerational wealth moments from the same PSID linked sample; all wealth in thousands of 2016 dollars. Entries marked "—" await model output from the full 35-moment calibration. See Appendix A for sample construction details.

Figure 3 complements Panel D by comparing the full wealth distribution in the model and the data. The model reproduces the right-skewed shape of the empirical wealth distribution and captures the concentration of wealth among higher-income households. The close fit of the wealth distribution is important because parent wealth is a key state variable governing both the college decision and the transfer policy: errors in the wealth distribution would propagate into misspecified college gradients and transfer flows.

Figure 3. Wealth Distribution: Model vs. Data

Notes: Comparison of the parent wealth distribution in the model (simulated) and the data (PSID). Wealth is measured in thousands of 2016 dollars.

6.2 The Role of Parental Altruism in College Attainment

To quantify the importance of parental altruism, I solve the model with $\eta = 0$: parents are not altruistic and make no transfers. In this economy, children must self-finance college through earnings, loans, and grants alone. Without altruism, college attendance no longer depends on parent wealth. Low-ability children from rich and poor families attend at nearly identical rates, because the wealth gradient in the baseline is entirely driven by parental transfers and the commitment channel. The effect of shutting down altruism is concentrated among low-ability children: attendance falls by 43%. High-ability children attend college regardless of parental support, because the private return to college is sufficiently high. These results confirm that parental altruism operates at the margin where the college decision is sensitive to financial support.

The mechanism works through two channels. First, altruistic parents directly subsidize college costs, reducing the net price their children face. The transfer amount increases with parent wealth but decreases with child ability, because parents use transfers to influence marginal children—those whose college decision depends on financial support. Second, parents internalize that a child without college will generate larger lifetime transfer costs. For wealthy parents, the present value of expected transfers to a high-school child exceeds the cost of college tuition, making college the more efficient investment. However, the child's perspective introduces an additional force: anticipated parental transfers during and after college subsidize enrollment, while anticipated lifetime transfers also cushion children who forgo college against income risk. The net effect on college attendance is therefore theoretically ambiguous. To disentangle these forces—direct cost subsidies, the moral hazard reduction motive, and the transfer cushion for non-college children—I next introduce the full commitment benchmark and construct counterfactual economies that eliminate ongoing parent-child interaction.

7 Moral Hazard Decomposition

Parental altruism raises college attendance, but it also generates a fundamental tension: the Samaritan's dilemma. Because children anticipate transfers when their wealth is low, they under-save relative to autarky, imposing larger and more frequent transfer costs on parents. A central question is how this lack of commitment affects the college decision. The answer is theoretically ambiguous. On one hand, anticipated parental support during the college years effectively subsidizes enrollment, making college more attractive to the child. On the other hand, anticipated lifetime transfers cushion non-college children against income risk, reducing their incentive to invest in human capital. Whether college attendance is higher or lower under the Samaritan's dilemma than under full commitment depends on which force dominates—a quantitative question that the structural model is designed to answer. This section introduces the full commitment benchmark, formalizes both forces, constructs counterfactual economies to isolate them, and reports the quantitative decomposition.

7.1 Full Commitment Benchmark and Two Opposing Forces

To isolate the effects of the Samaritan's dilemma, I first characterize the efficient allocation under full commitment. Suppose a benevolent planner commits at $t = 0$ to a state-contingent transfer schedule $\{\tau(a_p, a_c, z, t)\}$ for the entire overlap period. Because both agents pool resources under a single plan, the problem reduces to a single-agent optimization over total household wealth $A \equiv a_p + a_c$:

$$\rho V^{\mathrm{FC}}(A, z, t) = \max_{c_{\mathrm{tot}}} \left\{ u_{\mathrm{FC}}(c_{\mathrm{tot}}) + V_A^{\mathrm{FC}} \left[rA + y_p(t) + y_c(z, t) - c_{\mathrm{tot}} \right] + \mathcal{L}_z V^{\mathrm{FC}} + V_t^{\mathrm{FC}} \right\} \quad (14)$$

Under CRRA preferences, the efficient allocation equates weighted marginal utilities, $u'(c_p) = \eta\, u'(c_c)$, so consumption shares are time-invariant: $c_c = \eta^{1/\gamma}\, c_p$. Total consumption $c_{\mathrm{tot}} = c_p(1 + \eta^{1/\gamma})$ yields an aggregated flow utility $u_{\mathrm{FC}}(c_{\mathrm{tot}}) = \frac{c_{\mathrm{tot}}^{1-\gamma}}{1-\gamma}(1 + \eta^{1/\gamma})^{\gamma}$, and the planner's HJB reduces to a single-state-variable problem. Full commitment differs from the baseline economy in three ways (Barczyk and Kredler, 2014a,b): it eliminates overconsumption by both agents, it renders the timing of transfers indeterminate, and—most important for the college decision—it provides insurance without moral hazard. The child's college choice under full commitment follows the same logit structure as the baseline, with the planner's continuation values replacing the baseline values.

Relative to full commitment, the Samaritan's dilemma creates two opposing forces on the child's college decision. From the parent's perspective, a college-educated child is cheaper to support: college permanently raises the income level and lowers income risk ($\sigma_C \ll \sigma_{HS}$), shifting the child away from the transfer boundary. Under full commitment, the parent could internalize this motive directly by conditioning transfers on e_c. In the baseline economy, however, the parent cannot commit: the transfer policy responds only to the current state (a_p, a_c, z), and the parent transfers whenever the child is sufficiently poor, regardless of the child's education choice.

The child chooses education to maximize own continuation utility, taking the parent's equilibrium transfer policy as given. The continuation values V_C and V_{HS} embed the entire expected path of future transfers. The child does not internalize the cost transfers impose on the parent, generating overconsumption relative to the efficient allocation whenever the child's state is near or inside $\mathcal{T}$. Two forces emerge from this interaction:

Force 1: Anticipated transfers subsidize enrollment. The child starts college with zero assets and reduced income ($\ell_C \cdot y_c$ net of tuition), placing the initial state deep in the transfer region $\mathcal{T}$. The parent—bound by time consistency—optimally transfers resources that effectively subsidize enrollment costs. This implicit subsidy raises the child's valuation of the college path relative to full commitment, pushing attendance *above* the efficient level.

Force 2: Anticipated transfers cushion non-college children. Children who forgo college face lower expected income and higher income risk ($\sigma_{HS} \gg \sigma_C$), so they spend substantially more of their adult lives in the transfer region. The parent cannot withhold this support. The anticipation of this lifetime safety net raises the child's valuation of the high-school path—the child can under-save and over-consume, knowing that parental transfers will materialize when income falls. This cushion reduces the child's incentive to attend college, pushing attendance *below* the efficient level.

Whether Force 1 or Force 2 dominates depends on ability, parental wealth, and the income process parameters. The decomposition below quantifies these forces and confirms that the sign of the moral hazard effect on college varies across the ability–wealth distribution.

7.2 Counterfactual Economies and Quantitative Decomposition

To isolate the role of commitment from the role of insurance, I construct two counterfactual economies that eliminate ongoing parent-child interaction entirely, replacing the dynamic transfer game with a one-time lump-sum transfer at $t = 0$. After the initial transfer, the child solves the standard lifecycle problem alone—shutting down both the Samaritan's dilemma and the parental insurance channel. The two counterfactuals differ in whether the parent gives the same transfer in both education states or an education-contingent menu (τ_C, τ_{HS}), which separates the pure moral hazard channel from the education-targeting channel.[7]

Counterfactual A: Unconditional one-time transfer. The parent chooses τ_0 at $t = 0$ and the child receives it regardless of education choice:

$$\max_{\tau_0 \in [0, a_p]} \quad V_p^{\text{alone}}(a_p - \tau_0) \\ + \eta \Big[\Pr(C \mid \tau_0, \theta)\, V^{c,\text{alone}}(\tau_0 \mid C) + \big(1 - \Pr(C \mid \tau_0, \theta)\big)\, V^{c,\text{alone}}(\tau_0 \mid HS) \Big] \quad (15)$$

where $V_p^{\text{alone}}(a)$ is the parent's remaining lifetime utility with no further interaction with the child. The child's education choice follows a logit:

$$\Pr(C \mid \tau_0, \theta) = \int \frac{\exp\big[\big(V^{c,\text{alone}}(\tau_0 \mid C) - \kappa(\theta, \varepsilon)\big)/\sigma_{cd}\big]}{\sum_{e \in \{C, HS\}} \exp\big[\big(V^{c,\text{alone}}(\tau_0 \mid e) - \kappa_e(\theta, \varepsilon)\big)/\sigma_{cd}\big]}\, \phi(\varepsilon)\, d\varepsilon \quad (16)$$

where $\kappa_C(\theta, \varepsilon) = \kappa(\theta, \varepsilon)$ and $\kappa_{HS}(\theta, \varepsilon) = 0$. Since both education paths receive the same endowment τ_0, the education decision reflects only fundamentals—human capital returns net of psychic cost—and not the structure of ongoing parental support. Comparing baseline attendance with this economy isolates the *pure moral hazard effect*: the distortion to college enrollment caused solely by the child's anticipation of future transfers.

Counterfactual B: Education-contingent transfer menu. The parent commits at $t = 0$ to a transfer schedule that may differ by the child's education choice, choosing a transfer τ_C paid if the child attends college and a transfer τ_{HS} paid if the child chooses high school:

$$\max_{(\tau_C, \tau_{HS}) \in [0, a_p]^2} \; \Pr(C \mid \tau_C, \tau_{HS}, \theta) \big[V_p^{\text{alone}}(a_p - \tau_C) + \eta\, V^{c,\text{alone}}(\tau_C \mid C) \big] \\ + \big[1 - \Pr(C \mid \tau_C, \tau_{HS}, \theta)\big] \big[V_p^{\text{alone}}(a_p - \tau_{HS}) + \eta\, V^{c,\text{alone}}(\tau_{HS} \mid HS) \big] \quad (17)$$

$$\Pr(C \mid \tau_C, \tau_{HS}, \theta) = \int \frac{\exp\big[\big(V^{c,\text{alone}}(\tau_C \mid C) - \kappa(\theta, \varepsilon)\big)/\sigma_{cd}\big]}{\exp\big[\big(V^{c,\text{alone}}(\tau_C \mid C) - \kappa(\theta, \varepsilon)\big)/\sigma_{cd}\big] + \exp\big[V^{c,\text{alone}}(\tau_{HS} \mid HS)/\sigma_{cd}\big]}\, \phi(\varepsilon)\, d\varepsilon \quad (18)$$

[7] Appendix H provides a formal derivation of the full commitment benchmark and discusses its relationship to the one-time transfer counterfactuals. The key distinction is that the FC benchmark removes moral hazard while preserving insurance, whereas the one-time transfer counterfactuals remove both.

where $\kappa(\theta, \varepsilon) \geq 0$ is the psychic cost of college (zero for the HS alternative). This menu is the efficient one-time-transfer benchmark: it nests Counterfactual A (the restriction $\tau_C = \tau_{HS}$) and a college-or-nothing economy (the restriction $\tau_{HS} = 0$), and because it relaxes both restrictions it weakly dominates them in parent welfare. Crucially, the altruistic parent can now support a child in the high-school state directly, so a low-return child is no longer pushed into college merely to capture parental resources. Comparing Counterfactual B with Counterfactual A isolates the *education-targeting effect*: the change in attendance when the parent can direct transfers by education rather than give the same endowment in both states.

Decomposition. Define:

$$\Delta_{\text{MH}}^{\text{pure}}(\theta, a_p) = \Pr(\text{College} \mid \text{baseline}) - \Pr(\text{College} \mid \text{unconditional}) \tag{19}$$

$$\Delta_{\text{targ}}(\theta, a_p) = \Pr(\text{College} \mid \text{unconditional}) - \Pr(\text{College} \mid \text{menu}) \tag{20}$$

$$\Delta_{\text{MH}}^{\text{total}}(\theta, a_p) = \Delta_{\text{MH}}^{\text{pure}} + \Delta_{\text{targ}} \tag{21}$$

A positive $\Delta_{\text{MH}}^{\text{pure}}$ indicates that the baseline economy raises college attendance relative to the no-moral-hazard economy; a negative value indicates that moral hazard depresses college attendance. The education-targeting component Δ_{targ} isolates the change in enrollment when the parent can direct transfers by education: a positive sign indicates that targeting *lowers* attendance, because the parent diverts low-return children out of college and supports them in the high-school state instead.

Results. Table 15 reports the decomposition.

Table 15. Moral Hazard Decomposition of College Attendance

	Overall	Parent Wealth Quartile			
		Q1 (Low)	Q2	Q3	Q4 (High)
Panel A: College Attendance Rate (all ability types)					
Baseline	0.385	0.368	0.374	0.388	0.409
No MH (unconditional)	0.341	0.430	0.364	0.382	0.186
No MH (menu)	0.246	0.395	0.341	0.127	0.122
$\Delta^{\text{pure}}_{\text{MH}}$	+0.044	−0.062	+0.010	+0.006	+0.224
$\Delta^{\text{total}}_{\text{MH}}$	+0.139	−0.027	+0.033	+0.261	+0.288
Panel B1: Pure Moral Hazard Effect (Δ^{pure}_{MH}) by Ability					
Low ability		−0.091	−0.104	−0.092	−0.058
Medium ability		−0.073	−0.067	−0.066	+0.430
High ability		+0.009	+0.009	+0.010	+0.526
Panel B2: Education-Targeting Effect (Δ_{targ}) by Ability					
Low ability		+0.179	+0.200	+0.205	+0.205
Medium ability		+0.000	+0.033	+0.474	+0.000
High ability		+0.012	+0.000	+0.048	−0.543
Panel B3: Total Effect ($\Delta^{total}_{MH} = \Delta^{pure}_{MH} + \Delta_{targ}$) by Ability					
Low ability		+0.088	+0.096	+0.113	+0.147
Medium ability		−0.073	−0.034	+0.407	+0.430
High ability		+0.021	+0.009	+0.058	−0.017
Panel C: Optimal One-Time Transfer τ_0^* ($000s)					
Unconditional:					
Low ability		126.2	295.4	654.1	1582.3
Medium ability		144.6	143.0	763.1	1360.8
High ability		144.6	143.0	130.8	1012.7
Menu, if college (τ_C):					
Low ability		0.0	0.0	0.0	0.0
Medium ability		144.6	143.0	1242.8	1297.5
High ability		31.5	143.0	130.8	1993.7
Menu, if high school (τ_{HS}):					
Low ability		144.6	295.4	675.9	1582.3
Medium ability		144.6	295.4	675.9	1487.3
High ability		144.6	143.0	872.1	1107.6

Notes: Panel A reports average college attendance across ability types under three economies: the baseline, the no-MH economy with an unconditional transfer, and the no-MH economy with the education-contingent menu (τ_C, τ_{HS}). The "Overall" column averages across wealth quartiles. In both no-MH economies, parent and child interact only at $t = 0$; the child solves the lifecycle problem alone thereafter.

Aggregate result. In the baseline economy, 38.5% of children attend college. Eliminating ongoing parent-child interaction and replacing it with an unconditional lump-sum transfer raises attendance to 43.9%—a pure moral hazard effect of $\Delta_{\text{MH}}^{\text{pure}} = -5.5$ percentage points (the anticipated transfer cushion of Force 2 depresses the baseline below the no-moral-hazard level). Allowing the parent to direct transfers by education—the menu (τ_C, τ_{HS})—then *lowers* attendance to 37.5%, an education-targeting effect of $\Delta_{\text{targ}} = +6.4$ percentage points: when the parent can support a child in the high-school state, the marginal low-return children who would have attended college only to capture parental resources now stay out and receive τ_{HS} instead. The two channels roughly offset, so the total effect of the Samaritan's dilemma on the college *level* is small ($\Delta_{\text{MH}}^{\text{total}} = +1.0$ pp). The economically meaningful distortion is therefore one of composition rather than level: relative to the efficient menu the baseline delivers about the right amount of college in aggregate but the wrong students, an interpretation the welfare analysis in Section 8 confirms. This also reverses the sign of the channel obtained under a college-or-nothing restriction ($\tau_{HS} = 0$), which mechanically forces attendance upward by withholding all support from non-college children.

Decomposition by ability. The aggregate result masks heterogeneity across the ability distribution; Panels B1–B3 of Table 15 disaggregate the effects by child ability. The targeting channel operates with opposite signs across ability. For low- and medium-ability children the menu sets the college premium $\tau_C - \tau_{HS}$ near zero or supports the high-school state directly ($\tau_{HS} > 0$), so the child stays out of college: the parent prefers to transfer resources rather than finance a degree whose return does not cover the human-capital gain net of the psychic cost. For high-ability children the menu sets $\tau_C > \tau_{HS}$, raising attendance, because the return and the reduction in future transfer costs justify a college-contingent premium. The pure moral hazard channel (Panel B1) is moderate and negative for low- and medium-ability children and turns positive for high-ability children, reflecting the enrollment-subsidy force (Force 1). Panel C reports the implied menu transfers (τ_C, τ_{HS}); in cells where no child attends in equilibrium the college-state transfer τ_C is off the equilibrium path and not separately identified.

7.3 How College Reshapes the Transfer Region

The decomposition identifies the quantitative importance of moral hazard; the mechanism operates through the transfer region $\mathcal{T}$. College reduces the child's exposure to the transfer region through two channels: a higher ability gradient ($\lambda_C = 0.797 > \lambda_{HS} = 0.517$) that permanently raises expected income, and substantially lower income risk ($\sigma_C \ll \sigma_{HS}$) that reduces the magnitude of negative shocks. Both forces keep college children's wealth further from the transfer boundary.

Figure 4 plots the fraction of simulated individuals in the transfer region at each age, by education. High-school children spend a substantially larger fraction of their adult life receiving transfers; college children exit the region earlier and return less frequently after negative shocks. Panel C of Table 16 quantifies this exposure, reporting the fraction of time receiving transfers separately for the in-college and post-college periods.

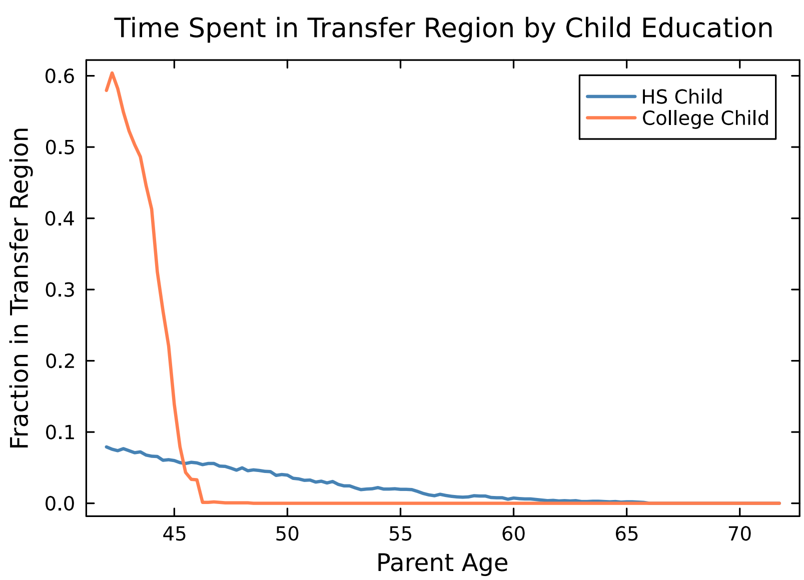


Figure 4. Fraction of Simulated Individuals in the Transfer Region, by Education

Notes: Fraction of simulated individuals ($M = 4{,}000$) receiving positive transfers ($\tau > 0$) at each parent age, by child education. High-school children spend more time in the transfer region throughout the overlap period.

Table 16. Transfer Region Exposure and Savings Behavior

	HS Child	College Child
Panel A: Child's own savings effort $s_c^{own} = ra_c + y_c - c_c$ *($k/yr)*		
When receiving transfers	0.24	-0.82
When not receiving transfers	0.84	1.84
Difference (Samaritan's dilemma)	-0.60	-2.65
Panel B: Over-consumption vs. autarky $\Delta c_c = c_c^{MPE} - c_c^{aut}$ *($k/yr)*		
In transfer region	-34.13	-13.65
Outside transfer region	-6.57	-21.11
Panel C: Fraction of time receiving parental transfers		
During college (ages 18–22)	—	0.362
After college (ages 22–48)	0.016	0.000

Notes: Statistics from simulated dynasties ($M = 4{,}000$). Panel A reports the child's own savings effort $s_c^{\text{own}} = \tilde{r}_c\, a_c + y_c - c_c$, which excludes the mechanical effect of transfers on wealth accumulation. A negative difference indicates the child saves less of their own resources when receiving transfers (the Samaritan's dilemma). Panel B reports the consumption differential relative to an overlap-period autarky benchmark c_c^{aut}: the child's optimal consumption from a lifecycle problem solved at the same age and wealth level with no parent interaction, using the simulated parent wealth to compute the EFC for financial aid. A negative value indicates the child in the baseline economy consumes less than the autarky benchmark at the same state. Panel C reports the fraction of time the child receives parental transfers, separately for the in-college years (ages 18–22, the enrollment-subsidy channel) and the post-college period (ages 22–48, the insurance channel); the in-college figure applies only to college children.

The reduced transfer-region exposure translates into lower lifetime transfer flows. Figure 5 shows average lifetime transfers by parent wealth quartile and child education. Across all quartiles, college children receive substantially fewer transfers, with reductions ranging from 30% to 60%. This reflects both the mechanical effect (higher income reduces the need for support) and the behavioral effect (reduced moral hazard induces more self-insurance). The key insight is that a dollar spent on tuition generates a *permanent* increase in the child's income drift, reducing the probability of entering the transfer region for the remainder of the lifecycle. A dollar transferred directly provides temporary consumption smoothing but does not change the drift, so the child returns to the transfer boundary over time.

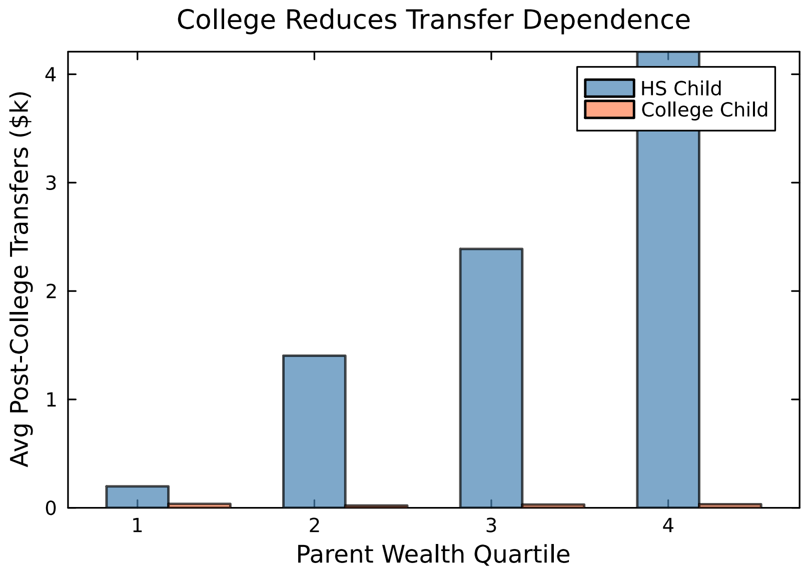


Figure 5. Lifetime Transfers by Parent Wealth and Child Education

Notes: Average post-college transfers received by the child (excluding the college period), by parent wealth quartile and child education. Percentages indicate the transfer reduction from college relative to high school. Source: simulated dynasties ($M = 4{,}000$).

8 Welfare Analysis

The preceding section established that the Samaritan's dilemma distorts both college enrollment and savings behavior. This section quantifies the welfare costs of these distortions and evaluates a free-college policy in the presence and absence of commitment.

8.1 Consumption-Equivalent Variation

I measure welfare using the consumption-equivalent variation (CEV): the permanent proportional change in consumption that makes an agent indifferent between the baseline (Samaritan's dilemma) economy and an alternative. Under CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, CRRA homogeneity implies that scaling all consumption flows by a factor $(1 + \Delta)$ scales lifetime utility by $(1 + \Delta)^{1-\gamma}$. Inverting gives the CEV

$$\Delta = \left(\frac{V^{\text{alt}}}{V^{\text{base}}}\right)^{\frac{1}{1-\gamma}} - 1, \tag{22}$$

where $\Delta > 0$ means the alternative is welfare-improving.

For the child, $V^c = \int_0^T u(c_c)\, e^{-\rho t}\, dt$ and equation (22) applies directly. For the parent, $V^p = \int_0^T [u(c_p) + \eta\, u(c_c)]\, e^{-\rho t}\, dt$ captures both own and altruistic consumption. Because u is homogeneous of degree $1-\gamma$, scaling both c_p and c_c by $(1+\Delta)$ scales V^p by $(1+\Delta)^{1-\gamma}$. The parent CEV therefore measures the permanent household-wide consumption change—own plus the altruistic component—that achieves indifference.

I compute CEV at each grid point (a_p, θ) by evaluating the value functions at the beginning of the overlap period ($a_c = 0$, $z = 0$, $t = 0$), integrating over the logit college-choice shock. I report unweighted averages across the (a_p, θ) grid unless stated otherwise.

8.2 Welfare Across Commitment Structures

Table 17 reports CEV for three alternative economies relative to the baseline. The first two rows of Panel A correspond to the no-moral-hazard counterfactuals from Section 7: the unconditional one-time transfer eliminates ongoing transfers but gives the child the same endowment in both education states, while the education-contingent menu lets the parent choose separate transfers (τ_C, τ_{HS}). Both allocations remove the Samaritan's dilemma by construction—the parent commits to a transfer schedule at $t = 0$ and the child solves an independent problem thereafter.

Measured by the dynastic (η-weighted) consumption-equivalent variation—the welfare object the parent optimizes—the economies rank in order: the unconditional transfer (32.8% relative to baseline) below the menu (39.7%) below full commitment (50.8%). Two forces generate this ordering. The menu weakly dominates the unconditional transfer because the latter is the restricted case $\tau_C = \tau_{HS}$; the gain is a targeting gain, not a coercion gain, as the parent matches transfers to each child's optimal education—paying a college premium where the return is high and supporting low-return children directly in the high-school state—rather than raising enrollment to capture the human-capital return. Full commitment dominates both one-time-transfer economies because it removes moral hazard while *preserving* insurance, whereas the one-time transfers remove both. The parent-own and child-own columns of Table 17 show that the menu's dynastic gain accrues mainly to the child: relative to the unconditional transfer, the parent transfers more efficiently toward the child, raising the child's own value while leaving the parent's own value roughly unchanged. This corrects the interpretation under a college-or-nothing benchmark, where the apparent welfare gain came partly from mechanically pushing low-return children into college.

Although the parent ranks these commitment economies above the baseline and values the child's utility directly, it cannot implement them, and the reason is precisely the parent's altruism. Committing to hand the child a one-time transfer and then stand back is not time-consistent. Ex ante the parent would like to promise to transfer τ_0 once and never again, inducing the child to self-insure and eliminating the moral hazard. Ex post, however, whenever the child draws a bad shock and approaches the transfer region, the parent—precisely because $\eta > 0$—strictly prefers to transfer rather than let the child consume little. The child anticipates this and under-saves, so the promise unravels: the only time-consistent (Markov-perfect) policy is the baseline one, in which the parent transfers whenever the child is sufficiently poor, regardless of education. The commitment allocations are therefore benchmarks the parent would adopt if a commitment technology existed, not policies it can choose unilaterally—the "freedom" the child enjoys in the no-moral-hazard economies is something the altruistic parent cannot credibly grant with cash. This is exactly why college matters: by permanently raising the child's income and lowering income risk, a tuition payment moves the child away from the transfer region for the remainder of life, achieving through human capital the commitment that direct transfers cannot.

8.3 Free-College Policy

I consider a policy that eliminates tuition ($\phi = 0$) and finances the revenue shortfall through a lump-sum tax T^{LS} on initial parent wealth. No financial aid is needed when tuition is zero, so $\bar{g} = 0$. The tax is set in general equilibrium to balance the government budget: it must cover tuition for the children who enroll, while enrollment itself responds to the tax through the parent's post-tax wealth. The per-dynasty tax therefore solves the fixed point

$$T^{\mathrm{LS}} = 4\phi \cdot \mathbb{E}\big[\Pr\big(\mathrm{C} \mid a_p - T^{\mathrm{LS}},\, \theta\big)\big]\,, \tag{23}$$

where ϕ is the annual tuition defined above, 4ϕ is the four-year cost of tuition the government covers per enrolled child, and the expectation is over the (a_p, θ) distribution. Enrollment, transfers, and value functions are all evaluated at the post-tax wealth $a_p - T^{\text{LS}}$. Because enrollment differs across the two behavioral regimes, the budget-balancing tax is solved separately for each. Parents with $a_p < T^{\text{LS}}$ have post-tax wealth clamped at zero; they still benefit because their child pays no tuition.

I solve the free-college economy under two regimes. Under *no commitment*, the model retains its Markov-perfect equilibrium structure—parents and children interact strategically with $\phi = 0$—and the Samaritan's dilemma persists. Under *commitment*, the parent commits to a one-time education-contingent menu (τ_C, τ_{HS}) at $t = 0$ and the child solves independently with $\phi = 0$.

Panel B of Table 17 reports the results. Under no commitment, free college raises enrollment from 35.2% to 39.2% but delivers a dynastic CEV gain of only 2.0%: the subsidy is largely dissipated because the child continues to anticipate ongoing support and under-saves. Under commitment, the parent redirects the tuition saving into the education-contingent menu; enrollment *falls* to 29.6% as low-return children are channeled toward direct support rather than a subsidized degree, and the dynastic CEV gain rises to 36.9%. The value of free college thus comes not from raising enrollment but from letting the parent—when able to commit—reallocate the freed resources efficiently across education states.

Panel C reports the share of the free-college welfare gain that is dissipated by the lack of commitment, computed as $1 - \Delta^{\text{no cmt}}/\Delta^{\text{cmt}}$, where Δ denotes the CEV gain from free college over the baseline under each regime. Since Δ^{cmt} reflects free college together with the move to commitment, this is the fraction of the achievable gain that is forfeited when parents cannot commit. In the estimated model 94.7% of the dynastic gain is dissipated, and the share rises steeply with parental wealth—from roughly one-fifth in the bottom wealth quartile to over 95% in the top—because wealthier families rely more heavily on transfers and therefore suffer more from the commitment problem.

9 Conclusion

This paper studies why altruistic parents invest in their children's college education and how this investment interacts with the moral hazard costs of intergenerational transfers. I develop a dynastic model in which college serves as a commitment device that reduces the

Table 17. Welfare Analysis: CEV (%) Relative to Baseline

	Parent (own) CEV (%)	Child CEV (%)	Dynastic ($+\eta$) CEV (%)
Panel A: Commitment Structure			
Baseline (Samaritan's dilemma)	0.00	0.00	0.00
Unconditional transfer (no MH)	0.42	105.20	32.78
Education-contingent menu (no MH)	-0.51	140.64	39.72
Full commitment (planner)	21.02	101.34	50.77
Panel B: Free College Policy ($\phi = 0$, lump-sum tax)			
Free college, no commitment	0.25	4.35	1.95
Free college, with commitment	-0.30	125.38	36.93
Panel C: Share of free-college gain dissipated by lack of commitment			
Share (%)		96.5	

Notes: Consumption-equivalent variation (CEV) measures the permanent proportional change in consumption that achieves indifference between each economy and the baseline. The *Parent (own)* column uses the parent's own-consumption value V_p^{own}; the *Child* column uses the child's own value; the *Dynastic* column uses the η-weighted dynastic value $V_p^{\text{own}} + \eta V_c^{\text{own}}$ that the parent optimizes and is the welfare-relevant aggregate. Panel A compares commitment structures holding tuition at its estimated value. Panel B sets $\phi = 0$ and funds the policy with a budget-balancing lump-sum tax T^{LS} (equation 23) solved in general equilibrium for each regime. Panel C reports the share of the free-college welfare gain dissipated by the lack of commitment, $1 - \Delta^{\text{no cmt}}/\Delta^{\text{cmt}}$. Source: model solution at estimated parameters.

Samaritan's dilemma, and I test its predictions using matched parent-child data. A set of stylized facts—parent consumption is higher when children attend college, wealth paths diverge by child college status, college attenuates dynastic precautionary saving, and college reduces the probability of transfers while leaving conditional transfer amounts unchanged or larger—are consistent with the model's central mechanism: college shrinks the transfer region, releasing precautionary resources.

The net effect of the Samaritan's dilemma on college attendance is theoretically ambiguous—the lack of commitment creates an enrollment subsidy (Force 1) that raises attendance and a transfer cushion (Force 2) that depresses it. The structural decomposition shows that these forces roughly offset on the aggregate enrollment *level*, so the economically important distortion is one of *composition*: relative to an efficient benchmark in which the parent can target transfers by education, the lack of commitment leaves marginal low-return children enrolled while under-subsidizing high-return ones. The enrollment subsidy dominates for high-ability children—for whom anticipated transfers raise attendance—while low- and medium-ability children are diverted out of college and supported directly. The misallocation carries a welfare cost that grows with parental wealth, and it sharply attenuates the gains from a free-college policy when parents cannot commit. The same altruism parameter η that generates the Samaritan's dilemma also drives parents to invest in college as a way to mitigate it.

The framework carries implications for the design of college subsidies and financial aid. Policies that reduce college costs benefit both students and their parents, who face lower lifetime transfer obligations. However, the model suggests that the welfare effects of financial aid depend on how policies interact with private transfers: grants that crowd out parental contributions may be less effective at increasing enrollment than those that complement family support. Conversely, policies that expand parents' capacity to transfer without encouraging education may exacerbate the Samaritan's dilemma by reducing children's incentives to invest in their own human capital.

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A PSID Sample Construction

This appendix describes in detail the construction of the analysis sample from the Panel Study of Income Dynamics (PSID). The sample draws on four distinct PSID data products: the main Family Interview File, the Individual File, the Transition into Adulthood Supplement (TAS), and the Family Identification Mapping System (FIMS). I use 13 biennial waves covering 1999–2023.

A.1 Data Sources

Main Family Interview File. The core source is the PSID family-level interview file (data extract J361818), which contains 3,611 variables and 33,440 family-year observations. This file provides household-level information on demographics, income, wealth, employment, and housing tenure for each survey wave from 1999 to 2023.

Supplement File. A second family-level file (data extract J361820, 132 variables, 18,890 observations) provides supplementary variables—primarily industry codes (3-digit Census, 2003–2015) and savings—that are not available in the main extract due to PSID's modular data structure.

Individual File. The PSID individual file (data extract J361819, 85,536 individuals) provides person-level information including age, relationship to household head, sequence number within the family unit, and—beginning in 2013—years of completed education for each family member.

Transition into Adulthood Supplement (TAS). The TAS (data extract J361814) covers young adults aged approximately 18–28 in the 2015 and 2017 waves. I use it to directly identify college attendance via variables `TA150731` ("Whether in college now," 2015 wave) and `TA170790` ("Whether attending/attended college," 2017 wave).

Family Identification Mapping System (FIMS). FIMS provides the intergenerational linkage between children and their parents. Each child record contains identifiers for up to four parent types: biological father, biological mother, adoptive father, and adoptive mother. Parent identifiers consist of the 1968 Interview Number (`ER30001`) and Person Number (`ER30002`), which together form a time-invariant person identifier.

A.2 Variable Construction

All variables are harmonized across the 13 waves (1999–2023) using PSID variable codes (ER-series). Due to the Stata/IC limit of 2,048 variables, I use a selective fixed-width import strategy that reads approximately 350 variables per wave from the main file. Table A.1 provides the complete variable mapping.

Table A.1. PSID Variables Used in Analysis

Variable	Description	Waves	Source
Panel A: Demographics			
`AgeRef`	Age of reference person (head)	1999–2023	Family File
`AgeSpouse`	Age of spouse/partner	1999–2023	Family File
`NumChildren`	Number of children in family unit	1999–2023	Family File
`AgeYoungChild`	Age of youngest child	1999–2023	Family File
`stateCode`	State of residence (FIPS)	1999–2023	Family File
`raceHead`	Race of reference person	1999–2023	Family File
`educHead`	Education of reference person	1999–2023	Family File
`educSpouse`	Education of spouse	1999–2023	Family File
Panel B: Income			
`totalFamilyIncome`	Total family income ($)	1999–2023	Family File
`headLaborIncome`	Head's labor income ($)	1999–2023	Family File
`spouseLaborIncome`	Spouse's labor income ($)	1999–2023	Family File
Panel C: Wealth (PSID Wealth Supplement)			
`TotalWealth`	Total net worth incl. home equity	1999–2023	Wealth Module
`WealthNonHouse`	Net worth excl. home equity	1999–2023	Wealth Module
`HomeEquity`	Home value minus mortgage	1999–2023	Wealth Module
`AnnuityIRA`	Annuity and IRA accounts	1999–2023	Wealth Module
`savings`	Savings accounts	2003–2023	Supplement
`stocks`	Stocks/mutual funds/inv. trusts	1999–2023	Wealth Module
`Vehicles`	Net vehicle value	1999–2023	Wealth Module
`OtherAssets`	Other assets	1999–2023	Wealth Module
`OtherHome`	Other real estate	1999–2023	Family File
`studentLoans`	Student loan debt	2011–2023	Family File
`otherDebts`	Other debts	2013–2023	Family File
Panel D: Employment			
`emplStatus`	Employment status of head	1999–2023	Family File
`workingNow`	Currently working (binary)	1999–2023	Family File
`reasonLeftJob`	Reason left last job	1999–2023	Family File
`industryCode`	Census industry (3d 03–15; 4d 17+)	2003–2023	Supplement
`occupationCode`	Census occupation (3d 03–15; 4d 17+)	2003–2023	Family/Supp
Panel E: Consumption (separate extract)			
`foodExpenditure`	Total food expenditure	1999–2023	Family File
`housingExpenditure`	Total housing expenditure	1999–2023	Family File
`transportExpenditure`	Total transportation expenditure	1999–2023	Family File
`healthExpenditure`	Total health care expenditure	1999–2023	Family File
`clothingExpenditure`	Total clothing expenditure	2005–2023	Family File
`recreationExpenditure`	Recreation expenditure	2003–2023	Family File
Panel F: Individual-Level (from Individual File)			
`uniqueID`	Person ID ($\texttt{ER30001} \times 1000 + \texttt{ER30002}$)	—	Indiv. File
`seqNum`	Sequence number in family unit	1999–2023	Indiv. File
`relToHead`	Relationship to reference person	1999–2023	Indiv. File
`age`	Age of individual	1999–2023	Indiv. File
`yearEduc`	Years of completed education	2013–2023	Indiv. File

Notes: All monetary variables deflated to 2016 dollars using the CPI-U. Wealth variables for 1999–2007 use PSID-imputed "S-series" codes; from 2009 onward, wealth is in the main family file. The `savings` variable is unavailable prior to 2003 and is obtained from the supplement file. Industry and occupation codes switch from 3- to 4-digit classification in 2017.

A.3 Sample Construction Pipeline

The sample is constructed in six sequential steps, each corresponding to a separate script in the replication package.

A.3.1 Step 1: Import and Reshape Family File

The main family file is imported using a selective fixed-width (`infix`) command that reads only the approximately 350 variable-columns needed for the analysis. For each of the 13 waves, I extract the wave-specific variables, rename them to standardized names (e.g., `ER13010` → `AgeRef` for 1999, `ER82018` → `AgeRef` for 2023), and append all waves into a single long panel. The supplement file (J361820) is processed identically and merged on `InterviewID` × `year`, filling in variables that are only available in the supplement (principally `savings` and detailed industry codes for 2003–2015).

I harmonize industry and occupation codes across the classification break in 2017:

$$\texttt{industryCode} = \begin{cases} \texttt{industry3d} & \text{if } 2003 \leq \texttt{year} \leq 2015 \\ \texttt{industry4d} & \text{if } 2017 \leq \texttt{year} \leq 2023 \end{cases}$$

A 2-digit industry variable ($\texttt{industry2d} = \lfloor \texttt{industryCode}/10 \rfloor$) is available for analyses requiring consistency across all waves. An involuntary separation indicator is defined as $\texttt{reasonLeftJob} \in \{1, 2\}$ (plant closing or layoff).

The output is a family-year panel (`FamFileSubSetStata.dta`) with one observation per family-interview per wave.

A.3.2 Step 2: Import and Reshape Individual File

The individual file provides person-level data for all PSID sample members. I import 85,536 individuals using a selective `infix` that reads the following per wave: interview number (`InterviewID`), sequence number (`seqNum`), relationship to reference person (`relToHead`), age, and—from 2013 onward—years of completed education (`yearEduc`). The data are reshaped into a person × year panel. A time-invariant person identifier is constructed as:

$$\texttt{uniqueID} = \texttt{ER30001} \times 1000 + \texttt{ER30002}$$

where `ER30001` is the 1968 family lineage number and `ER30002` is the person number within that lineage. This identifier is used throughout for parent–child linkage.

A.3.3 Step 3: Construct Household Panel

The family-year panel and individual-year panel are merged on `InterviewID` × `year`. The resulting dataset is restricted to current family unit members (sequence number 1–20) who are either the household head (`relToHead` = 10), the legal spouse (`relToHead` = 20), or a cohabiting partner (`relToHead` = 22). Including spouses ensures that parents who are not the household head—typically mothers in two-parent households—can be matched to their children via FIMS. Duplicate person-year observations (rare edge cases) are resolved by keeping the first occurrence, with preference given to "head" status. Each person's `uniqueID` is stored as `parent_uniqueID` for the subsequent intergenerational merge.

A.3.4 Step 4: Import Intergenerational Links (FIMS)

The FIMS file maps each PSID sample member to their parents. For each child, up to four parent identifiers are available:

- Biological father (`ER30001_P_F`, `ER30002_P_F`)
- Biological mother (`ER30001_P_M`, `ER30002_P_M`)
- Adoptive father (`ER30001_P_AF`, `ER30002_P_AF`)
- Adoptive mother (`ER30001_P_AM`, `ER30002_P_AM`)

FIMS also provides a generation identifier (`GID`): GID = 1 for 1968 original sample members, GID = 2 for their children, GID = 3 for grandchildren, and so on. The target sample for this paper consists primarily of GID ≥ 3 children (born approximately 1975–2005, reaching college age between 1993 and 2023) matched to their GID ≥ 2 parents.

A.3.5 Step 5: Import Transfer and Aid Supplement (TAS)

The TAS covers young adults in the 2015 and 2017 waves. I construct a `uniqueID` for each TAS respondent using the same formula as for the individual file. The key variable is college attendance status: `ta150731` $= 1$ indicates current college enrollment in 2015, and `ta170790` $= 1$ indicates college attendance in 2017. Respondents who refuse to answer (`ta170790` $= 9$) are dropped.

A.3.6 Step 6: Construct Analysis Sample

The final analysis sample is constructed by linking children to their parents' household data. The procedure is as follows:

(a) Identify college attendance. College enrollment is identified using a hybrid approach that combines three methods to maximize coverage across all 13 waves:

1. **Direct observation (2013–2023):** A child is coded as attending college if $12 <$ `yearEduc` < 17 and $18 \leq$ `age` ≤ 24. Additionally, an increase in `yearEduc` between consecutive waves signals active enrollment for individuals with `yearEduc` > 12 and `age` ≤ 26.

2. **Age-based proxy (1999–2011):** Since `yearEduc` is unavailable before 2013, children aged 18–22 whose maximum observed education (from later waves) exceeds 12 years are coded as attending college.

3. **TAS confirmation (2015, 2017):** Direct indicators from the TAS supplement override the above for the two available waves. If TAS explicitly reports non-enrollment (`ta150731` $= 5$ or `ta170790` $= 5$), the college indicator is set to zero.

(b) Link children to parents. Each child is linked to a parent household using FIMS identifiers. The parent is selected using the following priority hierarchy:

$$\text{biological father} \rightarrow \text{adoptive father} \rightarrow \text{biological mother} \rightarrow \text{adoptive mother}$$

The parent's unique identifier is computed as `parent_uniqueID` = `ER30001_P` $\times$ 1000 + `ER30002_P`, where the suffix is determined by the priority above. Children with no matchable parent in any category are dropped.

(c) Merge with parent household data. The child–parent linked file is merged with the household panel (`hh_data.dta`) on `parent_uniqueID` $\times$ `year`. Only matched observations are retained.

(d) Collapse to parent-household level. Since the unit of analysis is the parent household, I collapse the data to one observation per `parent_uniqueID` $\times$ `year`. At this stage, I count the number of children currently in college (`NumkidsCollege`) per parent-household-year.

(e) Merge tuition data. I merge state-level college cost data:

- In-state public tuition (by `stateCode` $\times$ `year`), constructed from IPEDS.
- Private college tuition (by `stateCode` $\times$ `year`).
- National average private tuition (by `year`).

All monetary variables are deflated to 2016 dollars using the CPI-U.

(f) Define income groups. Permanent income is computed as the average real household income observed when the reference person is aged 26–59 (i.e., strictly between 25 and 60). For parents not observed in this age range during the sample, I use their overall average income as a fallback. Income groups are defined as:

$$\begin{aligned} &\text{Group 1:} && \$30{,}000 < \bar{y} < \$60{,}000 \\ &\text{Group 2:} && \$60{,}000 \leq \bar{y} < \$100{,}000 \\ &\text{Group 3:} && \$100{,}000 \leq \bar{y} < \$160{,}000 \\ &\text{Group 4:} && \bar{y} \geq \$160{,}000 \end{aligned}$$

Observations with average income outside these ranges or with missing income are dropped.

A.4 Sample Restrictions

The following observations are dropped from the analysis sample:

1. Missing or invalid demographic data: `AgeRef` = 999 or `AgeSpouse` = 999.
2. Missing or invalid education: `educHead` = 9 or `educSpouse` = 9.
3. Top-coded or missing wealth/income: any of `totalFamilyIncome`, `WealthNonHouse`, `AnnuityIRA`, `studentLoans`, `savings`, `OtherDebts`, `TotalWealth`, `Vehicles`, `OtherHome`, or `stocks` equal to 9,999,999 or 9,999,998 (PSID missing/refused codes).
4. Negative total wealth: `TotalWealth` < 0.
5. Logical inconsistencies in portfolio: `TotalWealth` $<$ `HomeEquity` or `TotalWealth` $<$ `OtherHome`.
6. Extreme portfolio shares: home equity share $< -40\%$, other real estate share $< -40\%$, or vehicle share $> 50\%$ of total wealth.
7. Income outside classification bounds: average real income $\leq$ \$30,000 or unclassifiable.

A.5 Variable Definitions

Household consumption is the sum of expenditures on food at home, food away from home, housing (rent or imputed rent for homeowners), utilities, transportation, health care, education, and child care. From 2005 onward, clothing, recreation, and vacation are also included. Income is total household income before taxes. Wealth is total net worth, defined as the sum of home equity, other real estate, vehicles, farm and business assets, stocks, checking and savings accounts, bonds, IRAs, and other assets, minus all debts. Inter-vivos transfers are reported separately for transfers from parents to children and transfers from children to parents, and include cash gifts, help with expenses, and regular financial support. The key explanatory variable throughout the empirical analysis is child college attainment: an indicator for whether the child holds a bachelor's degree or higher. To condition on parent resources, I construct within-cohort parent income quartiles, ranking each parent relative to others born in the same year.

A.6 Key Constructed Variables

Portfolio shares. Asset portfolio shares are computed as percentages of total net worth:

$$\text{IRA share} = (\texttt{AnnuityIRA}/\texttt{TotalWealth}) \times 100$$
$$\text{Home equity share} = (\texttt{HomeEquity}/\texttt{TotalWealth}) \times 100$$
$$\text{Other real estate share} = (\texttt{OtherHome}/\texttt{TotalWealth}) \times 100$$
$$\text{Vehicle share} = (\texttt{Vehicles}/\texttt{TotalWealth}) \times 100$$

Instrumental variable. The college cost instrument interacts tuition with the number of children currently in college:

$$\texttt{InStateCollCostbykids}_{it} = \log(\text{InStateTuition}_{s(i),t}) \times \texttt{NumkidsCollege}_{it}$$

where $s(i)$ denotes parent i's state of residence.

Involuntary job separation. An indicator for involuntary separation is defined as:

$$\texttt{involuntary\_sep}_{it} = \mathbf{1}[\texttt{reasonLeftJob}_{it} \in \{1, 2\}]$$

where code 1 denotes plant closing and code 2 denotes layoff/termination.

Real variables. All monetary variables are deflated to 2016 dollars using the CPI-U All Urban Consumers index. Log transformations are applied where noted in the regression specifications.

A.7 Consumption Data

PSID consumption expenditure data are available for all waves from 1999 onward. The consumption measure aggregates expenditures across six categories: food (at home and away), housing (rent, utilities, property taxes, insurance), transportation (vehicle costs, fuel, parking, public transit), health care (insurance premiums, out-of-pocket), clothing (available 2005+), and recreation (available 2003+). These variables are extracted in a separate `infix` pass due to Stata/IC variable limits and merged to the main panel on `InterviewID` × `year`.

A.8 Summary of PSID Data Files

Table A.2 summarizes the raw PSID data files used in this paper.

Table A.2. PSID Raw Data Files

File	**Extract ID**	**Content**	**Size**
J361818.txt	Main family file	Demographics, income, wealth, employment	3,611 vars
J361820.txt	Supplement	Industry codes, savings, additional vars	132 vars
J361819.txt	Individual file	Person-level panel (all HH members)	85,536 persons
J361814.txt	TAS	College attendance for young adults	44 vars
FIMS_Data.csv	FIMS	Parent–child intergenerational links	All persons
InStateCollgeCost.dta	—	IPEDS in-state tuition by state × year	External
PrivateCollegeCost.dta	—	IPEDS private tuition by state × year	External
cpi.dta	—	CPI-U deflator (base 2016)	BLS

Notes: PSID data are from the 2023 release (waves 1999–2023, biennial). All PSID data are publicly available from the PSID Data Center (`https://psidonline.isr.umich.edu`). Tuition data are from IPEDS. CPI data are from the Bureau of Labor Statistics.

A.9 Merge Attrition

Table A.3 documents the sample size at each stage of the data pipeline: starting from the full PSID individual file, linking children to parents via FIMS, and merging with household-level data on consumption, income, and wealth. Panel B breaks down the parent–child links by relationship type (biological father, adoptive father, biological mother). Panel C quantifies the observations lost at each merge stage. The main source of attrition is the requirement that the identified parent appears as a household head or spouse in the PSID family file during the sample period.

Table A.3. PSID Sample Construction and Merge Attrition

Stage	Observations	Unique IDs	% Retained
Panel A: Building the Analysis Sample			
Individual file (all persons × years)	441 158	45 386	—
After college identification	441 152	45 386	100.0
Matched to FIMS (child–parent link)	358 874	34 916	81.3
Matched to parent household (hh_data)	148 541	10 492	41.4
Collapsed to parent-HH × year	71 896	10 492	—
After sample restrictions	62 320	10 145	86.7
Panel B: Parent Identification Source			
Linked via biological father	297 284		
Linked via adoptive father	4114		
Linked via biological mother	56 856		
No identifiable parent (dropped)	0		
Panel C: Merge Losses			
Individuals not in FIMS	82 278		
Child–years without parent in hh_data	210 333		
Parent–years without state tuition data	16 488		
Final analysis sample	62 320	10 145	
of which: HH–years with child in college	6566		

Notes: Children are linked to parents via FIMS. The parent–child link follows a hierarchical priority: biological father, adoptive father, biological mother, adoptive mother. "Unique IDs" refers to unique persons (first two rows) or unique parent households (remaining rows). "% Retained" is relative to the previous stage. All monetary variables winsorized at the 1st and 99th percentiles and deflated to 2016 dollars.

A.10 PSID Variable Codes by Wave

Table A.4 provides the complete mapping of standardized variable names to PSID ER-codes for each wave. This mapping is necessary for replication, as PSID assigns unique variable codes to each concept in each wave.

Table A.4. Selected PSID ER-Codes by Wave (Family File)

Variable	**1999**	**2001**	**2003**	**2005**	**2007**	**2009**	**2011**	**2013**	**2015**	**2017**	**2019**	**2021**	**2023**
Interview ID	ER13002	ER17002	ER21002	ER25002	ER36002	ER42002	ER47302	ER53002	ER60002	ER66002	ER72002	ER78002	ER82002
Age Head	ER13010	ER17013	ER21017	ER25017	ER36017	ER42017	ER47317	ER53017	ER60017	ER66017	ER72017	ER78017	ER82018
Total Income	ER16462	ER20456	ER24099	ER28037	ER41027	ER46935	ER52343	ER58152	ER65349	ER71426	ER77448	ER81775	ER85629
Head Labor Inc.	ER16463	ER20443	ER24116	ER27931	ER40921	ER46829	ER52237	ER58038	ER65216	ER71293	ER77315	ER81642	ER85496
Spouse Labor Inc.	ER16465	ER20447	ER24135	ER27943	ER40933	ER46841	ER52249	ER58050	ER65244	ER71321	ER77343	ER81670	ER85524
Total Wealth	S417	S517	S617	S717	S817	ER46970	ER52394	ER58211	ER65408	ER71485	ER77511	ER81838	ER85692
Home Equity	S420	S520	S620	S720	S820	ER46966	ER52390	ER58207	ER65404	ER71481	ER77507	ER81834	ER85688
AnnuityIRA	S419	S519	S619	S719	S819	ER46964	ER52368	ER58181	ER65378	ER71455	ER77481	ER81808	ER85662
Stocks	S411	S511	S611	S711	S811	ER46954	ER52358	ER58171	ER65368	ER71445	ER77471	ER81798	ER85652
Vehicles	S413	S513	S613	S713	S813	ER46956	ER52360	ER58173	ER65370	ER71447	ER77473	ER81800	ER85654
Empl. Status	ER13205	ER17216	ER21123	ER25104	ER36109	ER42140	ER47448	ER53148	ER60163	ER66164	ER72164	ER78167	ER82150
Educ. Head	ER15951	ER20012	ER23449	ER27416	ER40588	ER46566	ER51927	ER57683	ER64835	ER70907	ER76922	ER81169	ER85146
Race Head	ER15928	ER19989	ER23426	ER27393	ER40565	ER46543	ER51904	ER57659	ER64810	ER70882	ER76897	ER81144	ER85121

Notes: The S-series codes (S411–S820) refer to PSID-imputed wealth supplement variables used in waves 1999–2007; from 2009 onward, wealth variables are recorded directly in the main family file with ER-codes. The complete set of codes used in this paper covers approximately 350 variables across all 13 waves. The full mapping is available in the replication package.

A.11 Individual File Variables by Wave

Table A.5 documents the individual-file variable codes used to construct the person-year panel.

Table A.5. Individual File ER-Codes by Wave

Variable	**1999**	**2001**	**2003**	**2005**	**2007**	**2009**	**2011**	**2013**	**2015**	**2017**	**2019**	**2021**	**2023**
Interview ID	ER33501	ER33601	ER33701	ER33801	ER33901	ER34001	ER34101	ER34201	ER34301	ER34501	ER34701	ER34901	ER35101
Sequence No.	ER33502	ER33602	ER33702	ER33802	ER33902	ER34002	ER34102	ER34202	ER34302	ER34502	ER34702	ER34902	ER35102
Rel. to Head	ER33503	ER33603	ER33703	ER33803	ER33903	ER34003	ER34103	ER34203	ER34303	ER34503	ER34703	ER34903	ER35103
Age	ER33504	ER33604	ER33704	ER33804	ER33904	ER34004	ER34104	ER34204	ER34305	ER34504	ER34704	ER34904	ER35104
Years Educ.	—	—	—	—	—	—	—	ER34230	ER34349	ER34548	ER34752	ER34952	ER35152

Notes: Years of completed education (`yearEduc`) is only available in the Individual File from 2013 onward. For earlier waves, college attendance is identified using the age-based proxy described in Section A.3. The time-invariant person identifiers `ER30001` (1968 Interview Number, bytes 2–5) and `ER30002` (Person Number, bytes 6–8) are constant across all waves.

B NLSY97 Sample Construction

This appendix describes the construction of the NLSY97 analysis sample. The National Longitudinal Survey of Youth 1997 (NLSY97) tracks a nationally representative cohort of 8,984 Americans born between 1980 and 1984, interviewed annually from 1997 through 2017 (biennially after 2011). I use the NLSY97 for two purposes: constructing college graduation rates by ability and parent wealth quartile (the 16 moments used in SMM estimation), and estimating the education-specific income process.

B.1 Data Sources

Main Survey File. The raw NLSY97 download contains 2,257 variables across all survey rounds. Because the Stata/IC variable limit of 2,048 is binding, I use a Python preprocessing script (`00_subset_nlsy97.py`) to extract the approximately 110 variables needed for the analysis before loading into Stata.

Schooling Module (YSCH Series). The YSCH-24600 series records detailed college enrollment and financing information at the college$\times$term$\times$year level. Variables include tuition charges, room and board costs, grants and scholarships received, total loans, out-of-pocket expenses, and parental transfers. These are reshaped from the NLS Investigator's wide format into a college–term–year panel.

Income and Transfer Module (YINC Series). The YINC-5800 and YINC-8200 variables record total parental transfers and parental allowances, respectively, for each survey year from 1997 to 2003. These capture financial support from parents during the college-age years.

Wage and Hours Data. The YINC-1700 series provides annual labor income, and the CVC_HOURS_WK_YR series provides annual hours worked, used to estimate the education-specific income process.

B.2 Variable Construction

Table B.1 summarizes the key variables used from the NLSY97.

Table B.1. NLSY97 Variables Used in Analysis

Variable	Description	Years	Source
Panel A: Demographics and Ability			
`PUBID_1997`	Individual identifier	—	Survey ID
`KEYSEX_1997`	Gender	1997	Baseline
`KEYRACE_ETHNICITY`	Race/ethnicity (4 categories)	1997	Baseline
`KEYBDATE_Y_1997`	Birth year	1997	Baseline
`ASVAB_MATH_VERBAL`	ASVAB math-verbal percentile	1999	CAT-ASVAB
`CVC_SAT_VERBAL`	SAT verbal score	2007 XRND	Transcript
`CVC_SAT_MATH`	SAT math score	2007 XRND	Transcript
`CVC_ACT_SCORE`	ACT composite score	2007 XRND	Transcript
`TRANS_CRD_GPA`	Overall high school GPA	—	Transcript
Panel B: Family Background			
`CV_HGC_BIO_MOM`	Bio. mother's highest grade	1997	Baseline
`CV_HGC_BIO_DAD`	Bio. father's highest grade	1997	Baseline
`CV_INCOME_GROSS_YR`	Gross family income	1997–2003	Annual
`CV_CENSUS_REGION`	Census region at baseline	1997	Baseline
Panel C: College Enrollment and Costs (YSCH-24600)			
Tuition (`YSCH_21900`)	Tuition charged per term	1997–2017	Schooling
Grants (`YSCH_25400`)	Grants and scholarships	1997–2017	Schooling
Loans (`YSCH_25600`)	Total student loans	1997–2011	Schooling
Out-of-pocket (`YSCH_24600`)	Out-of-pocket college expenses	1998–2001	Schooling
Room & board	Room and board costs	1998–2001	Schooling
Credit hours	Credit hours attempted	1997–2017	Schooling
Panel D: Education Attainment			
`YSCH_2857`	Highest grade attended	1998–2003	Annual
`CVC_HIGHEST_DEGREE`	Highest degree ever received	XRND	Cumulative
College withdrawal	Whether withdrew from college	Annual	Schooling
Panel E: Income Process			
`YINC_1700`	Annual labor income	1997–2011	Income
`CVC_HOURS_WK_YR`	Annual hours worked (all jobs)	1997–2007	Employment
Panel F: Parental Transfers			
`YINC_5800`	Total parental transfers ($)	1997–2003	Income
`YINC_8200`	Parental allowance ($)	1997–2003	Income

Notes: XRND denotes cross-round variables computed by NLS staff from the cumulative survey history. The ASVAB composite score is the math-verbal percentile from the CAT-ASVAB administered in 1999. The YSCH schooling variables are indexed by college enrollment number, term, and survey year. Parental income is gross household income of the respondent's family of origin.

B.3 Sample Construction

B.3.1 Step 1: Subset and Reshape

The raw NLSY97 file (2,257 variables) is first subset to approximately 110 variables using a Python script, then loaded into Stata. I keep core demographics (gender, race, birth year, census region), parental education (highest grade completed for biological mother and father), cognitive ability (ASVAB math-verbal percentile score from the 1999 administration), academic performance (SAT/ACT scores, high school GPA from transcripts), and family income from 1997 to 2003.

B.3.2 Step 2: College Enrollment Variables

The detailed schooling module variables (YSCH series) are reshaped from the NLS Investigator's wide format into long panels indexed by individual, college enrollment number, term, and year. Separate reshape scripts process each topic: tuition (`YSCH_21900`), grants and scholarships (`YSCH_25400`), total loans (`YSCH_25600`), out-of-pocket expenses (`YSCH_24600`), room and board, credit hours taken, and college withdrawal indicators. These are then merged into a unified college-spell panel.

B.3.3 Step 3: Education Attainment

College attendance is identified from the highest grade attended variable (`YSCH_2857`), available for 1998–2003. I define college attendance as the maximum grade attended exceeding 12 (i.e., any postsecondary enrollment). College graduation is defined as attaining a bachelor's degree by age 25, using the cumulative highest-degree variable (`CVC_HIGHEST_DEGREE_EVER`).

B.3.4 Step 4: Income Process Estimation

For the income process, I use annual labor income (`YINC_1700`) and hours worked (`CVC_HOURS_WK_YR_ALL`) to construct hourly wages by education group. Following Abbott et al. (2019), I estimate the high-school-to-college income ratio and education-specific income standard deviations, which provide 3 of the moments used in SMM estimation.

B.3.5 Step 5: Parental Transfers

Parental financial transfers (`YINC_5800`) and allowances (`YINC_8200`) are reshaped into a person-year panel for 1997–2003. These variables capture total financial support from parents during the college-age years and are used to calibrate the transfer structure in the model.

B.4 Key Constructed Variables

Ability quartiles. I rank individuals by their ASVAB math-verbal percentile score (administered in 1999) and partition into quartiles. The sample is restricted to individuals with non-missing ASVAB scores.

Parent wealth quartiles. Total household net worth at age 17 is used to construct within-cohort parent wealth quartiles. Together with ability quartiles, these define the 4×4 grid of college graduation rates used as the core estimation moments.

College graduation rates. The 16 cell-level graduation rates (ability quartile $\times$ parent wealth quartile) are the primary target moments for SMM estimation. These trace how college attainment varies across the joint distribution of ability and parental resources.

Average tuition. The gross annual cost of college (\$12,200) is computed from average tuition reported in the NLSY97 schooling module, following Abbott et al. (2019). The model uses a net (after institutional discounts) annual tuition of ϕ = \$6,700, with the remaining gap covered by the endogenous financial aid schedule $g(a_p)$ described in Section 3.7.

C HRS Sample Construction

This appendix describes the construction of the HRS analysis sample. The Health and Retirement Study (HRS) is a biennial longitudinal survey of Americans over age 50, conducted by the University of Michigan since 1992. I use the RAND HRS Longitudinal File (2022 version 1), which harmonizes variables across all waves (1992–2022), together with three supplemental files: the RAND Family-Kids file, the RAND Family-Resident file, the Consumption and Activities Mail Survey (CAMS), and the HRS Supplemental Survey on College Support (HUMS).

C.1 Data Sources

RAND HRS Longitudinal File. The core source is the RAND HRS 2022v1 longitudinal file, which contains 19,880 variables in wide format (one row per respondent, wave-varying variables indexed by wave number 1–16, corresponding to 1992–2022). The RAND staff harmonize variable definitions, handle skip patterns, and construct consistent wealth, income, and health variables. Due to the Stata/IC variable limit, I use a Python preprocessing script (`00_subset_hrs_wide.py`) to extract the approximately 860 needed variables before loading into Stata.

The RAND naming convention embeds the wave number: household-level variables use `h{wave}{stem}` (e.g., `h6atotb` = total wealth, wave 6), respondent-level variables use `r{wave}{stem}`, spouse-level variables use `s{wave}{stem}`, and time-invariant variables use `ra{stem}`.

RAND Family-Kids File. The family-kids file (`randhrsfamk1992_2014v1.dta`) records information about each respondent's children, including demographics (age, gender, education, marital status), geographic proximity, financial transfers in both directions, care provision, and bequest intentions. The file is in wide format with approximately 400 variables after subsetting, and is reshaped to a kid×wave panel.

RAND Family-Resident File. The family-resident file (`randhrsfamr1992_2014v1.dta`) provides household composition variables—number of living children, sons, and daughters—by wave. After subsetting to approximately 40 variables, this is reshaped to a respondent×wave panel.

CAMS (Consumption and Activities Mail Survey). CAMS (`randcams_2001_2017v1.dta`) is a supplemental mail survey that collects detailed consumption and spending data for a subset of HRS households, biennially from 2001 to 2017. Variables include total household spending, durable and non-durable spending, housing expenditure, and mortgage payments.

HUMS (Supplemental Survey on College Support). The HUMS supplement collects retrospective information on parents' financial contributions to their children's college education. For each child who attended college, the survey records tuition amounts, room and board costs, years of college, and the percentage of costs paid by the parent. I use HUMS to construct the total parental contribution to college in real 2016 dollars.

Exit Interview (Heritage File). The HRS exit interview, conducted with a proxy informant after a respondent's death, records bequest and inheritance information. I use the Heritage file to construct child-level bequest amounts (dollar transfers and estate percentages) and housing inheritance (whether each child inherited the main residence and the imputed value).

College Tuition Data. Regional in-state college tuition costs by census division and year (`InStateCollgeCost.csv`) are merged with HRS respondents using their census division of residence. These provide the regional college cost variation used in the analysis.

C.2 Variable Construction

Table C.1 summarizes the key variables constructed from the HRS data products.

Table C.1. HRS Variables Used in Analysis

Variable	**Description**	**Waves**	**Source**
Panel A: Respondent Demographics			
`AgeRespondant`	Age of respondent	1–16	RAND HRS
`AgeSpouse`	Age of spouse	1–16	RAND HRS
`RespondGender`	Gender	Time-inv.	RAND HRS
`RespondantRace`	Race	Time-inv.	RAND HRS
`RespondantYearEducation`	Years of education	Time-inv.	RAND HRS
`RespondantDegree`	Highest degree	Time-inv.	RAND HRS
`RespondantBirthYear`	Birth year	Time-inv.	RAND HRS
`SampleCohort`	HRS sample cohort	Time-inv.	RAND HRS
`RespondCensusDiv`	Census division	1–16	RAND HRS
Panel B: Household Wealth			
`TotalWealth`	Total net worth	1–16	RAND HRS
`NvalueHome`	Net home value	1–16	RAND HRS
`NvalueIRA`	IRA/Keogh accounts	1–16	RAND HRS
`Nsavings`	Checking/savings	1–16	RAND HRS
`Nvehicles`	Vehicle value	1–16	RAND HRS
`NvalueBusiness`	Business/farm assets	1–16	RAND HRS
`NotherRealStateAssets`	Other real estate	1–16	RAND HRS
Panel C: Income			
`TotalHouseholdIncome`	Total HH income	1–16	RAND HRS
`IncomeSSretirement`	SS retirement income	1–16	RAND HRS
`IncomeSSDisSSI`	SS disability/SSI	1–16	RAND HRS
`IncomePension`	Pension income	1–16	RAND HRS
Panel D: Bequest Expectations			
`ProbBequest10`	Prob. bequest $\geq$ \$10K	1–16	RAND HRS
`ProbBequest100`	Prob. bequest $\geq$ \$100K	1–16	RAND HRS
`ProbBequest500`	Prob. bequest $\geq$ \$500K	1–16	RAND HRS
`ProbAnyBequest`	Prob. any bequest	1–16	RAND HRS
Panel E: Child Information (Family-Kids File)			
`kidYearEducation`	Child's years of education	1–12	Family-Kids
`kidAge`	Child's age	1–12	Family-Kids
`kidGender`	Child's gender	Time-inv.	Family-Kids
`kidMarried`	Child's marital status	1–12	Family-Kids
`kidIncomeRange`	Child's income bracket	1–12	Family-Kids
`kid10miles`	Lives within 10 miles	1–12	Family-Kids
Panel F: Inter-Vivos Transfers (Family-Kids File)			
`kParentGiveFinancialTransfer`	Parent gave transfer (0/1)	1–12	Family-Kids
`kParentGiveAmount`	Amount parent gave ($)	1–12	Family-Kids
`kidAnyTransfer`	Child gave transfer (0/1)	1–12	Family-Kids
`kidTransferAmountImp`	Amount child gave ($, imp.)	1–12	Family-Kids
Panel G: College Support (HUMS)			
`totalParentContribution`	Tuition + board paid (2016$)	Retro.	HUMS
`NumKidsWentCollege`	N children in college	Retro.	HUMS
`ParentSupportKid`	Parent contributed >0	Retro.	HUMS
Panel H: Consumption (CAMS)			
`totalHouseholdConsumption`	Total HH consumption	2001–17	CAMS
`totalNonDurableSpenditure`	Non-durable spending	2001–17	CAMS
`totalHousingSpending`	Housing expenditure	2001–17	CAMS
Panel I: Bequests (Exit Interview)			
`amountTransferChild`	Dollar bequest to child	Exit	Heritage
`perEstateTransChild`	% of estate to child	Exit	Heritage
`AmountHouseInherited`	Housing inheritance value	Exit	Heritage

Notes: Waves 1–16 correspond to HRS survey years 1992–2022. The Family-Kids and Family-Resident files cover waves 1–12 (1992–2014). All monetary variables in the RAND HRS file are nominal; deflation to 2016 dollars occurs in the analysis scripts. HUMS variables are retrospective (covering the child's entire college period). Child income brackets in the Family-

C.3 Sample Construction Pipeline

C.3.1 Step 1: Import and Reshape RAND HRS Longitudinal File

The full RAND HRS file (19,880 variables, one row per respondent) is first subset to approximately 860 variables using a Python script (`00_subset_hrs_wide.py`). The subset retains time-invariant identifiers (`hhid`, `hhidpn`), demographic variables (birth year, gender, race, education, religion), and wave-varying variables for wealth (22 stubs), income (4 stubs), health (4 stubs), bequest expectations (4 stubs), family structure (4 stubs), employment (2 stubs), and spouse characteristics (10 stubs). The data are then reshaped from wide to long format using Stata's `reshape long` command, producing a person×wave panel.

Several variables require type conversions for downstream compatibility: `hhid` (string to numeric), `SampleCohort`, `RespondGender`, `RespondantYearEducation`, `RespondantDegree`, `RespondantRace`, `InterviewStatus`, and `IndustryCode` are converted from labeled numeric to string using `decode/tostring`, as downstream scripts compare against full label strings.

C.3.2 Step 2: Clean Family-Kids File

The Family-Kids file is subset and reshaped from wide (one row per child, waves as columns) to long (one row per child×wave). A unique child identifier is constructed as:

$$\texttt{uniqueKidID} = \texttt{hhidpn} \times 10^5 + \text{last 5 digits of } \texttt{kidid}$$

This links each child to their parent respondent. Variables are renamed to human-readable names and organized into demographic, child-to-parent transfer, parent-to-child transfer, and bequest intention groups.

C.3.3 Step 3: Clean Family-Resident File

The Family-Resident file provides household composition by wave: total number of living children, sons, and daughters. This is reshaped from wide to long and merged with the main panel to provide the child count breakdown.

C.3.4 Step 4: Consumption Data (CAMS)

The CAMS file is reshaped from wide to long format, yielding a respondent×wave panel with total household spending, durable and non-durable spending, housing expenditure, mortgage payments, and total household consumption. CAMS covers a subsample of HRS households biennially from 2001 to 2017.

C.3.5 Step 5: College Support (HUMS)

The HUMS supplement is cleaned and merged with the HUMS imputation file. For each child who attended college (`H3`=1), I compute:

$$\texttt{totalParentContribution} = \text{tuition}_{\text{real}} \times \text{years} \times \frac{H9}{10} + \text{board}_{\text{real}} \times \text{years}_{\text{board}} \times \frac{H11}{100}$$

where tuition and board are deflated to 2016 dollars using the CPI-U, $H9$ is the fraction of tuition paid (scale 0–10), and $H11$ is the fraction of room and board paid (scale 0–100). Observations with missing codes (`H5`=99, `H9`≥99.9, `H11`=999) are dropped; `H11`=997 ("other/inapplicable") is recoded to zero. The total family contribution sums across all children in the household.

C.3.6 Step 6: Spouse Death File

A spouse death file is constructed by extracting the death year of each HRS respondent and renaming their identifier to `SpouseIdentifier`. When merged with the main panel on `SpouseIdentifier`, this brings in the death year of the respondent's partner, enabling analysis of widowhood effects.

C.3.7 Step 7: Exit Interview and Bequests

The Heritage file from the exit interview provides two types of child-level bequest information:

Financial bequests. Dollar amounts and estate percentages transferred to each child are cleaned by removing HRS missing codes (999998, 999999, 9999998, 9999999, 99998, 99999, 9998, 9999 for amounts; 998, 999 for percentages). When the percentage is missing but dollar amounts are available, the percentage is imputed as each child's share of total transfers. Small amounts ($\leq$100) are treated as percentage values rather than dollar amounts. These are merged with the parent–child link file using `kidid`.

Housing inheritance. For respondents whose main residence was disposed to children (`dispositionMainHome` $\in \{2,3\}$) and not inherited by the spouse, I identify which children received the house using the OPN (Other Person Number) codes. When the code indicates "all children" (OPN = 993), the value is split equally. The per-child housing inheritance is:

$$\texttt{AmountHouseInherited} = \frac{\texttt{valueMainRes}}{N_{\text{recipients}}} \times \mathbf{1}[\text{child received house}]$$

C.3.8 Step 8: Regional College Cost Merge

Average in-state college tuition by census division and year is imported from an external CSV file and reshaped into a region×year panel. This is merged with HRS respondents using the census division of residence (`RespondCensusDiv`).

C.4 Sample Restrictions

I impose the following restrictions, analogous to those applied to the PSID sample:

1. **Age restrictions:** Parents over age 50 and children over age 26.

2. **Interview status:** Only respondents with completed interviews in each wave (`InterviewStatus` indicates a completed interview).

3. **Couple households:** For analyses requiring spouse information, I restrict to coupled households using the `CoupleHH` indicator.

4. **Missing data:** Observations with HRS missing codes (.d = don't know, .r = refused, .m = missing) are handled according to the specific analysis. For wealth and income, these codes are preserved through the data construction stage and handled in the analysis scripts.

The final HRS analysis sample contains 19,179 parent–child pairs and 98,861 observations.

D Income Process: Discrete-to-Continuous-Time Mapping and Life-Cycle Profiles

This appendix derives the mapping from the discrete-time AR(1) income process estimated by Abbott et al. (2019) to the continuous-time Ornstein-Uhlenbeck (OU) specification used in the model, and presents the implied life-cycle income profiles.

D.1 Mapping Derivation

The discrete-time income shock process is an AR(1) at annual frequency:

$$z^e_{t+1} = \rho^{\text{ann}}_e \, z^e_t + \eta^e_{t+1}, \qquad \eta^e_{t+1} \sim N\big(0,\, (\sigma^e_\eta)^2\big). \tag{24}$$

The continuous-time counterpart is the OU process:

$$dz^e = -\kappa_e \, z^e \, dt + \sigma_e \, dW, \tag{25}$$

where $\kappa_e > 0$ is the mean-reversion rate and σ_e is the diffusion coefficient.

Mean-reversion rate. The exact discretization of (25) at unit time intervals ($\Delta = 1$ year) gives the conditional mean $\mathbb{E}[z^e_{t+1} \mid z^e_t] = e^{-\kappa_e} z^e_t$. Matching this with the AR(1) coefficient ρ^{ann}_e yields:

$$e^{-\kappa_e} = \rho^{\text{ann}}_e \quad \Longrightarrow \quad \kappa_e = -\ln\big(\rho^{\text{ann}}_e\big). \tag{26}$$

Diffusion coefficient. The conditional variance of the discretized OU process is:

$$\text{Var}(z^e_{t+1} \mid z^e_t) = \frac{\sigma_e^2}{2\kappa_e}\big(1 - e^{-2\kappa_e}\big). \tag{27}$$

The conditional variance of the AR(1) is simply $(\sigma_\eta^e)^2$. Equating the two and using $e^{-\kappa_e} = \rho_e^{\text{ann}}$:

$$
\begin{aligned}
(\sigma_\eta^e)^2 &= \frac{\sigma_e^2}{2\kappa_e}\big(1 - (\rho_e^{\text{ann}})^2\big) \\
\Longrightarrow \quad \sigma_e &= \sigma_\eta^e \cdot \sqrt{\frac{2\kappa_e}{1 - (\rho_e^{\text{ann}})^2}}.
\end{aligned}
\tag{28}
$$

This mapping preserves both the autocorrelation and the stationary variance $\text{Var}(z^e) = \sigma_e^2/(2\kappa_e)$ of the process. Note that the stationary variance of the OU process equals the unconditional variance of the AR(1), $(\sigma_\eta^e)^2/(1 - (\rho_e^{\text{ann}})^2)$, confirming internal consistency.

Numerical values. Applying (26)–(28) to the male estimates from Abbott et al. (2019) (Table 3.2): for high-school graduates ($\rho_{HS}^{\text{ann}} = 0.952$, $\sigma_\eta^{HS} = 0.07$), the mapping yields $\kappa_{HS} = 0.049$ and $\sigma_{HS}^{\text{OU}} = 0.114$; for college graduates ($\rho_C^{\text{ann}} = 0.966$, $\sigma_\eta^C = 0.08$), it yields $\kappa_C = 0.035$ and $\sigma_C^{\text{OU}} = 0.109$. The mean-reversion rates are of similar magnitude. The key distinction for the model's mechanism comes from the estimated diffusion parameters σ_C and σ_{HS} (Section 5), which govern the magnitude of income shocks. The estimated values ($\sigma_C = 0.038 \ll \sigma_{HS} = 0.151$) imply substantially lower income risk for college graduates, keeping them further from the transfer boundary and reducing the parent's lifetime transfer costs.

D.2 Life-Cycle Profiles

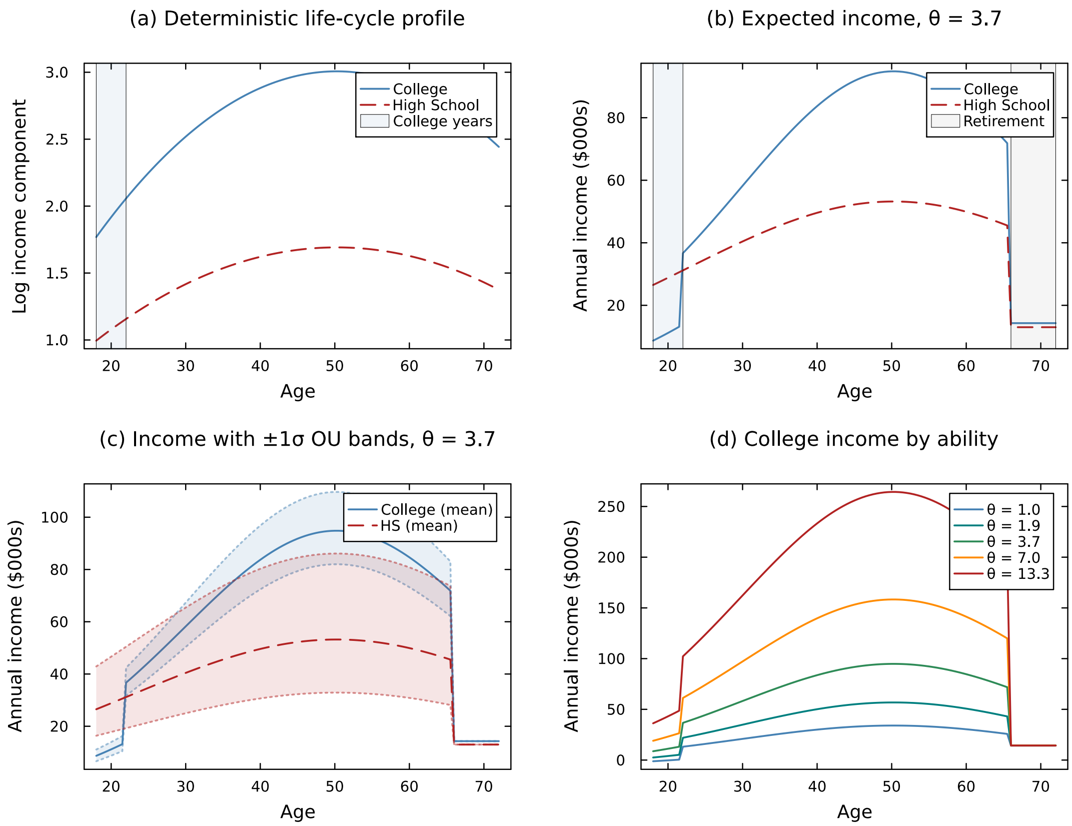

Figure D.1. Income Process: Life-Cycle Profiles by Education

Notes: Panel (a): deterministic age-income component $\gamma(t, e_c) = \gamma_{1,e}\, t - \gamma_{2,e}\, t^2$ from Abbott et al. (2019). Panel (b): expected annual income $y_c(t) = w \cdot \theta^{\lambda_e} \cdot \exp(\gamma(t, e_c))$ for median ability ($\theta = 3.9$); college workers earn $\ell_C = 56\%$ of full-time income during ages 18–22 and pay net tuition. Panel (c): ± 1 stationary standard deviation of the OU income shock ($\sigma_e/\sqrt{2\kappa_e}$) around expected income. Panel (d): college income profiles for low ($\theta = 1.5$), medium ($\theta = 3.9$), and high ($\theta = 10.2$) ability workers, illustrating the complementarity between ability and education ($\lambda_C > \lambda_{HS}$).

E Parent Wealth Portfolio

Table E.1 provides a detailed breakdown of parent income and wealth by income group. The table complements the summary variables reported in Table 1 in the main text.

Table E.1. Parent Income and Wealth by Income Group (PSID, 2016 Dollars)

	By Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
Household income	44,223	76,829	120,169	242,277	102,569
	(21,535)	(37,964)	(51,447)	(253,591)	(121,904)
Head labor income	23,523	40,976	64,728	123,504	53,930
	(19,369)	(28,162)	(42,902)	(123,970)	(64,202)
Total wealth	102,555	200,522	367,076	965,941	326,288
	(230,260)	(377,059)	(536,534)	(1,052,497)	(612,283)
Home equity	48,321	76,931	127,798	223,253	102,056
	(86,276)	(105,763)	(136,410)	(195,989)	(138,176)
IRA/annuity	6,437	20,026	49,226	107,888	35,668
	(38,692)	(68,967)	(108,878)	(163,964)	(98,914)
Non-housing fin. wealth	53,469	121,496	234,059	698,124	214,727
	(182,888)	(318,143)	(453,318)	(895,431)	(505,467)
Savings	7,950	16,013	36,321	68,743	25,676
	(31,182)	(39,546)	(61,142)	(87,368)	(56,070)
Vehicles	12,549	19,741	24,701	32,096	20,737
	(16,601)	(19,952)	(22,565)	(28,691)	(22,256)
Other real estate	7,938	13,934	22,799	55,426	20,513
	(42,810)	(57,200)	(75,043)	(122,965)	(73,634)
Student loans	1,736	4,134	7,932	7,755	4,881
	(7,821)	(12,628)	(18,488)	(20,246)	(14,759)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. All monetary variables deflated to 2016 dollars using the CPI-U and winsorized at the 1st and 99th percentiles. Income groups defined by parent average real household income observed between ages 25 and 60.

F Parent Consumption Detail

Table F.1 decomposes total household consumption into major expenditure categories. Housing constitutes the largest category across all income groups, followed by transportation and food. Discretionary categories—clothing, recreation, and vacation—are only available from the 2005 wave onward and increase sharply with income.

Table F.1. Parent Household Consumption by Income Group (PSID, 2016 Dollars)

	By Average HH Income Group				
	$30–60	$60–100	$100–160	$160+	All
Total consumption	36,426	46,131	57,243	73,130	50,076
	(18,374)	(20,389)	(24,489)	(33,277)	(26,252)
Food	7,915	9,211	10,582	12,872	9,718
	(4,854)	(4,944)	(5,296)	(6,269)	(5,470)
Housing	15,030	18,677	24,487	33,215	21,206
	(9,071)	(10,757)	(13,588)	(18,356)	(13,873)
Transportation	9,310	11,880	13,811	15,562	12,166
	(8,213)	(8,709)	(9,844)	(11,999)	(9,661)
Health care	2,108	3,402	4,195	5,027	3,469
	(3,027)	(3,649)	(4,157)	(4,832)	(3,950)
Clothing (2005+)	1,356	1,577	1,912	2,808	1,776
	(1,704)	(1,762)	(1,907)	(2,626)	(1,988)
Recreation (2005+)	503	807	1,178	1,805	957
	(920)	(1,211)	(1,464)	(1,897)	(1,396)
Vacation (2005+)	835	1,482	2,376	3,886	1,870
	(1,505)	(2,020)	(2,583)	(3,604)	(2,554)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. Consumption data available from the PSID beginning in 1999. All expenditure categories deflated to 2016 dollars and winsorized at the 1st and 99th percentiles. Clothing, recreation, and vacation are available from 2005 onward only. Total consumption is the sum of all listed categories.

G Additional Robustness: Parent Fixed Effects

As additional robustness for the consumption and wealth stylized facts (Sections 4.2 and 4.3), I exploit parent fixed effects to identify the college effect from within-family (between-sibling) variation. For consumption, the specification uses child-pair-year observations: within a given family and year, siblings who differ in college attainment generate variation in the college indicator while holding all parent-level characteristics constant.

Table G.1 reports parent fixed effects regressions of log parent consumption on the child's college indicator, at the child-pair-year level. The positive and significant coefficient confirms that the consumption difference documented in Section 4.2 is not driven by unobserved parental characteristics: even within the same family, parent consumption is higher in years when the college-educated child's characteristics are more salient.

Table G.1. Parent Consumption and Child College Status: Parent Fixed Effects (PSID)

	(1) OLS + Income Q FE	(2) Parent FE
College (BA+)	0.068***	0.027**
	(0.018)	(0.013)
Controls	Yes	Yes
Parent FE	No	Yes
Parent income Q FE	Yes	–
Observations	13,841	13,841
R^2 / Within R^2	0.469	0.840

Notes: Regressions of log parent household consumption on child college indicator (BA+), at the child-pair-year level. Column 1: OLS with parent income quartile fixed effects and comprehensive controls. Column 2: parent fixed effects with child age polynomial and education controls. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

H Full Commitment Benchmark

This appendix provides additional details on the full commitment benchmark introduced in Section 7.1 of the main text.

Under full commitment, a benevolent planner commits at $t = 0$ to a state-contingent transfer schedule $\{\tau(a_p, a_c, z, t)\}$ for the entire overlap period, solving a single-agent problem over total household wealth $A \equiv a_p + a_c$ (equation 14 in the main text). With CRRA preferences, the efficient allocation equates weighted marginal utilities:

$$u'(c_p) = \eta\, u'(c_c), \qquad \text{so that} \qquad c_c = \eta^{1/\gamma}\, c_p \tag{29}$$

Consumption shares are time-invariant. Substituting yields:

$$u_{\text{FC}}(c_{\text{tot}}) = \frac{c_{\text{tot}}^{1-\gamma}}{1-\gamma} \left(1 + \eta^{1/\gamma}\right)^{\gamma} \tag{30}$$

Relationship to the one-time transfer counterfactuals. The full commitment benchmark and the one-time transfer counterfactuals (Section 7.2) remove the Samaritan's dilemma through different mechanisms. Full commitment eliminates moral hazard while preserving state-contingent insurance: the planner adjusts transfers in response to income shocks throughout the overlap period. The one-time transfer counterfactuals eliminate both moral hazard and ongoing insurance: the child receives a lump sum at $t = 0$ and is subsequently on their own. The difference between FC and the one-time transfer economies therefore isolates the *insurance-removal effect*—the change in college attendance and welfare attributable to shutting down ongoing parental support, holding the absence of moral hazard fixed.

Comparing FC with the baseline economy yields the pure moral hazard effect holding insurance constant. As discussed in Section 7.1, this comparison reveals two opposing forces: anticipated transfers subsidize enrollment (raising college attendance above the efficient level) but also cushion non-college children (lowering attendance below the efficient level). The net sign depends on the child's position in the ability–wealth distribution.

I Equilibrium Properties

This appendix derives the key properties of the Markov-Perfect Equilibrium in the continuous-time coupled problem.

I.1 Optimality Conditions

The child's first-order condition from the HJB equation (2) gives $u'(c_c^*) = V_{a_c}^c$. In the no-transfer region ($\tau^* = 0$), the child's continuous-time Euler equation takes the standard form $\dot{c}_c/c_c = (r - \rho)/\gamma$. In the transfer region ($\tau^* > 0$), the transfer flow depends on the state, and the Euler equation becomes:

$$\frac{\dot{c}_c}{c_c} = \frac{1}{\gamma}\left(r - \rho + \frac{\partial \tau^*}{\partial a_c}\right) \tag{31}$$

The term $\partial\tau^*/\partial a_c < 0$ is the Samaritan's dilemma in continuous time: as child wealth increases, the parent reduces transfers, effectively lowering the child's return on saving below r and creating an incentive to over-consume.

The parent's first-order condition gives $u'(c_p^*) = V_{a_p}^p$. In the no-transfer region:

$$\frac{\dot{c}_p}{c_p} = \frac{1}{\gamma}(r - \rho) \tag{32}$$

In the transfer region, $V_{a_p}^p = V_{a_c}^p$: the parent equalizes marginal values of own and child wealth. The parent's wealth drift is exactly zero ($\dot{a}_p = 0$), as the parent channels all net resources to the child.

I.2 Transfer Region Characterization

The variational inequality (3) partitions the state space into:

1. **No-transfer region** $\mathcal{N} = \{(a_p, a_c, z, t) : V_{a_p}^p > V_{a_c}^p\}$: Parent and child accumulate wealth independently.

2. **Transfer region** $\mathcal{T} = \{(a_p, a_c, z, t) : V_{a_p}^p = V_{a_c}^p\}$: The transfer flow absorbs the parent's excess savings: $\tau^* = ra_p + y_p - c_p$.

The boundary is a free boundary determined endogenously. Parents transfer when the child is relatively poor (low a_c), faces negative income shocks (low z), or when the parent is wealthy (high a_p). The region shrinks as the parent approaches retirement ($t \to T^{\text{ret}}$).

I.3 The Samaritan's Dilemma

Both agents over-consume relative to the commitment solution. The child's distortion arises because anticipated transfers reduce the effective return on saving. Near the transfer boundary, the child anticipates that accumulating wealth will push the state into the no-transfer region, eliminating support. This creates a marginal disincentive to save, captured by $\partial\tau^*/\partial a_c$ in (31).

The parent's saving is also distorted: by saving more, the parent anticipates spending more time in the transfer region, which provides utility through child consumption but reduces the child's incentive to self-insure. In continuous time, this manifests as the parent's wealth drift being zero in $\mathcal{T}$.

The continuous-time formulation offers two advantages over discrete-time alternatives. First, transfers are a continuous flow rather than lump-sum payments, eliminating timing discreteness. Second, the free boundary provides a sharp characterization of where moral hazard operates.

I.4 College and Parental Influence

The college decision at $t = 0$ is forward-looking: the child compares lifetime values under each education path, incorporating the entire future trajectory of transfers. Altruistic parents affect the decision through two channels. First, they provide generous transfers during the college years (when child income is low), effectively subsidizing attendance—the transfer region $\mathcal{T}$ is larger when $e_c = C$ during the early years. Second, the continuation value under $e_c = C$ incorporates higher future income, which feeds back into the transfer policy: wealthier children require fewer transfers, freeing parent resources. The net effect generates a positive gradient of college enrollment in parent wealth.

J Solution Algorithm

The model is solved numerically using an upwind finite difference scheme with implicit time-stepping, following Achdou et al. (2022) and the transfer-region methods of Barczyk and Kredler (2014a,b). The algorithm proceeds in six stages: grid construction, solving the child-alone HJB, solving the coupled parent-child HJB system, computing college choice probabilities, forward simulation, and moment computation for SMM estimation.

J.1 State Space and Grid Construction

The model features three continuous state variables during the overlap period: parent wealth $a_p \in [0, \bar{a}_p]$, child wealth $a_c \in [\underline{a}_c, \bar{a}_c]$, and the persistent income shock $z \in [-\bar{z}, \bar{z}]$. In addition, child ability θ and education $e \in \{\text{HS}, \text{C}\}$ are discrete states that index separate HJB problems.

Asset grids. Both wealth grids use exponential spacing to concentrate points near the origin, where policy functions exhibit greater curvature. For N grid points from $a_{\min}$ to $a_{\max}$ with growth factor $g > 0$:

$$a_i = a_{\min} + a_{\max} \cdot \frac{(1+g)^{i-1} - 1}{(1+g)^{N-1} - 1}, \qquad i = 1, \dots, N. \tag{33}$$

The parent grid has $N_{a_p} = 30$ points over $[0, 4{,}000]$ (thousands of dollars). The child grid has $N_{a_c} = 30$ points, with approximately 20% of points allocated to the negative region $[\underline{a}_c, 0)$ using uniform spacing to accommodate student debt during college, and the remaining 80% exponentially spaced over $[0, \bar{a}_c]$. We set $\underline{a}_c = -25$ (the federal student borrowing limit) and $\bar{a}_c = 2{,}500$.

Income shock grid. The Ornstein-Uhlenbeck process $dz = -\kappa_e z \, dt + \sigma_e \, dW$ is discretized on a uniform grid of $N_z = 11$ points spanning ± 3 stationary standard deviations:

$$z_k \in \left\{ -3 \frac{\sigma_e}{\sqrt{2\kappa_e}}, \dots, +3 \frac{\sigma_e}{\sqrt{2\kappa_e}} \right\}, \qquad k = 1, \dots, N_z.$$

The grid width is set to the maximum stationary standard deviation across education types to ensure both HS and college processes are well-covered.

Ability grid. Child ability follows an AR(1) in logs across generations: $\log\theta' = \rho_\theta \log\theta + \mu_\theta(1-\rho_\theta)+\sigma_\theta\varepsilon$. We discretize this using the Tauchen (1986) method with $N_\theta = 5$ grid points spanning ± 3 stationary standard deviations of $\log\theta$.

Time grid. The backward induction uses $N_t = 120$ equally spaced time steps over the overlap period ($T^{\text{ret}} = 24$ years, child ages 18–42, parent ages 42–66) plus the parent's retirement period ($T - T^{\text{ret}} = 6$ years, parent ages 66–72), for a total horizon of $T = 30$ years, yielding $\Delta t = 0.25$ years per step. The coupled parent-child system is solved over $[0, T^{\text{ret}}]$; the parent's retirement phase over $(T^{\text{ret}}, T]$ is a single-agent problem embedded in the terminal condition.

J.2 Stage 1: Child-Alone HJB

After the overlap period ends (parent retires at age 66), the child solves a standard consumption-savings problem independently for the remainder of life ($T_{\text{child alone}} + T_{\text{child retire}} = 30$ years). The HJB equation is:

$$\rho V^{c,\text{alone}}(a,z,t) = \max_{c\geq 0}\left\{u(c) + V_a^{c,\text{alone}}\left[r\,a + y(t,z) - c\right] + V_z^{c,\text{alone}}(-\kappa z) + \tfrac{1}{2}\sigma^2 V_{zz}^{c,\text{alone}} + V_t^{c,\text{alone}}\right\}. \tag{34}$$

We solve this problem separately for each ability type θ and education level $e_c \in \{\text{HS}, \text{C}\}$, giving $2 \times N_\theta = 10$ independent HJB problems that are solved in parallel.

Terminal condition. At the child's death (age 72, i.e. $t = T_{\text{child}}$ on the child-alone clock), the terminal value is $V^c(a,z,T_{\text{child}}) = u(c_T)/\rho$ where $c_T = \max\{r\,a + \text{SS}_{e_c} + 0.1\,a,\ \epsilon\}$ represents consuming all remaining resources.

Optimal consumption via the first-order condition. At each grid point and time step, the FOC $u'(c^*) = V_a^c$ yields the candidate consumption $c^* = (V_a^c)^{-1/\gamma}$, where γ is the CRRA coefficient. We compute V_a^c using upwind finite differences: the *forward* difference $\partial_a^+ V$ and *backward* difference $\partial_a^- V$ yield two candidate consumption levels, c^{fwd} and c^{bwd}, with associated savings drifts $s^{\mathrm{fwd}} = ra + y - c^{\mathrm{fwd}}$ and $s^{\mathrm{bwd}} = ra + y - c^{\mathrm{bwd}}$. The upwind scheme selects:

$$c^* = \begin{cases} c^{\mathrm{fwd}} & \text{if } s^{\mathrm{fwd}} > 0 \quad \text{(saving: use forward difference),} \\ c^{\mathrm{bwd}} & \text{if } s^{\mathrm{bwd}} < 0 \quad \text{(dissaving: use backward difference),} \\ ra + y & \text{otherwise (drift} \approx 0\text{: consume resources).} \end{cases}$$

This upwind selection ensures numerical stability by using the derivative in the direction from which information flows.

Implicit time stepping. Given optimal policies at time step n, we construct the sparse infinitesimal generator $\mathbb{A}^n$ that encodes the upwind finite-difference operator in a and the discretized OU operator in z (combining upwind drift and central-difference diffusion terms). The implicit update solves:

$$\left[\left(\frac{1}{\Delta t} + \rho\right)\mathbf{I} - \mathbb{A}^n\right]\mathbf{V}^n = \mathbf{u}^n + \frac{1}{\Delta t}\mathbf{V}^{n+1}, \tag{35}$$

where $\mathbf{V}^n$ and $\mathbf{u}^n$ are the vectorized value function and flow utilities at step n, $\mathbf{I}$ is the identity matrix of dimension $N_{a_c} \times N_z$, and $\mathbf{V}^{n+1}$ is the value at the next (future) time step. This system is solved via sparse LU factorization.

OU generator. The infinitesimal generator for the OU process on the uniform z-grid combines a central-difference approximation for diffusion with upwind differencing for the drift:

$$\begin{aligned} (\mathbb{A}_z)_{k,k+1} &= \frac{\sigma^2/2}{\Delta z^2} + \frac{\max(-\kappa z_k, 0)}{\Delta z}, \\ (\mathbb{A}_z)_{k,k-1} &= \frac{\sigma^2/2}{\Delta z^2} + \frac{\max(\kappa z_k, 0)}{\Delta z}, \\ (\mathbb{A}_z)_{k,k} &= -(\mathbb{A}_z)_{k,k+1} - (\mathbb{A}_z)_{k,k-1}, \end{aligned} \tag{36}$$

with reflecting boundary conditions at $k = 1$ and $k = N_z$.

J.3 Stage 2: Coupled Parent-Child HJB

During the overlap period, the parent and child interact strategically. The parent's HJB equation is given by (1), with the transfer decision characterized by the complementarity condition (3). That is, the parent transfers whenever the marginal value of an additional dollar in the child's account exceeds the marginal value of retaining it.

Terminal condition. At $t = T^{\text{ret}}$ (parent age 66), the coupled system terminates. The parent continues into retirement until T (age 72) but no longer transfers to the child. The parent's terminal value combines a warm-glow term over remaining wealth and altruistic concern for the child's continuation value (equation 9). The child's terminal value is $V^c(a_p, a_c, z, T^{\text{ret}}) = V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p, z)$, the child-alone value function from Stage 1 evaluated at $t = 0$ with effective wealth including the bequest.

Policy iteration on the transfer region. We solve the coupled system backward in time. At each time step n, we iterate on the transfer region $\mathcal{T}^n \subseteq \{1, \ldots, N_{a_p}\} \times \{1, \ldots, N_{a_c}\} \times \{1, \ldots, N_z\}$ using the following variational inequality procedure, following Barczyk and Kredler (2014a):

1. **Initialize** the transfer region $\mathcal{T}^{n,0}$ (from the previous time step or empty).
2. **For** $\ell = 1, \ldots, L_{\max}$ (policy iterations):
   (a) **Compute parent consumption** from the FOC using upwind derivatives of V^p with respect to a_p:
   $$c_p = (V^p_{a_p})^{-1/\gamma},$$
   selecting the upwind direction based on the sign of the savings drift $s_p = ra_p + y_p - c_p$.
   (b) **Compute child consumption** from the FOC using upwind derivatives of V^c with respect to a_c:
   $$c_c = (V^c_{a_c})^{-1/\gamma}.$$
   (c) **Evaluate the transfer condition.** At each grid point (i, j, k), compare the parent's marginal values:

- If $\eta > 0$ and $V^p_{a_c}(i,j,k) > V^p_{a_p}(i,j,k)$: the parent benefits from transferring. Set parent consumption from the *child-wealth* marginal value, $c_p = (V^p_{a_c})^{-1/\gamma}$, and the transfer absorbs the parent's entire savings drift: $\tau = ra_p + y_p - c_p$. If $\tau > 0$, mark $(i,j,k) \in \mathcal{T}^{n,\ell}$.
- Otherwise: set $\tau = 0$ and $(i,j,k) \notin \mathcal{T}^{n,\ell}$.

(d) **Update wealth drifts:**

$$\dot{a}_p = \begin{cases} 0 & \text{if } (i,j,k) \in \mathcal{T}^{n,\ell}, \\ ra_p + y_p - c_p & \text{otherwise,} \end{cases} \qquad \dot{a}_c = \tilde{r}_c\, a_c + y_c - c_c + \tau.$$

In the transfer region, the parent channels all excess resources to the child, so $\dot{a}_p = 0$.

(e) **Check convergence:** if $\mathcal{T}^{n,\ell} = \mathcal{T}^{n,\ell-1}$, stop. Otherwise continue.

3. **Construct the generator and solve.** Given converged policies and drifts, assemble the sparse infinitesimal generator $\mathbb{A}^n$ over the full state space (a_p, a_c, z) of dimension $N_{a_p} \times N_{a_c} \times N_z$. The generator combines upwind operators in both wealth dimensions with the OU operator in z.

   A key computational insight is that the parent and child value functions share the *same* state transitions—both wealth drifts and the OU process are identical for both agents given the equilibrium policies. Therefore, the system matrix

$$\mathbf{B} = \left(\frac{1}{\Delta t} + \rho\right)\mathbf{I} - \mathbb{A}^n$$

   is factorized *once* (via sparse LU), and used for two triangular solves with different right-hand sides:

$$\mathbf{B}\,\mathbf{V}^{p,n} = \mathbf{u}^p + \frac{1}{\Delta t}\,\mathbf{V}^{p,n+1}, \tag{37}$$

$$\mathbf{B}\,\mathbf{V}^{c,n} = \mathbf{u}^c + \frac{1}{\Delta t}\,\mathbf{V}^{c,n+1}, \tag{38}$$

   where $\mathbf{u}^p = u(c_p) + \eta\, u(c_c)$ and $\mathbf{u}^c = u(c_c)$ are the flow utilities.

This procedure is repeated for each combination of child ability θ, child education $e_c \in \{\mathrm{HS}, \mathrm{C}\}$, and parent education $e_p \in \{\mathrm{HS}, \mathrm{C}\}$, yielding $N_\theta \times 2 \times 2 = 20$ coupled HJB problems that are solved in parallel across threads.

College-period modifications. When the child is in college (age < 22), two modifications apply: (i) the child earns a fraction ℓ_C of full-time income and pays net tuition $\phi_{\text{net}}(a_p) = \phi - \bar{g} \cdot \max(0,\, 1 - a_p/\bar{a}_{\text{efc}})$, which varies continuously with parent wealth through the EFC system; and (ii) the child may borrow up to $\bar{b}$ at the student loan rate $r + \iota_s$.

Borrowing constraints. The child's wealth drift is constrained to be non-negative at the borrowing limit: if $a_c \le \underline{a}_c + \epsilon$ and $\dot{a}_c < 0$, we set $\dot{a}_c = 0$. During college, $\underline{a}_c = -\bar{b}$; after college, $\underline{a}_c = 0$. The parent faces a natural borrowing constraint $a_p \ge 0$.

J.4 Stage 3: College Choice

At $t = 0$, the college enrollment decision combines the child's and parent's values with a paternalism weight $\psi \ge 0$. The probability of attending college follows the logit specification in equation (11), where all value functions are evaluated at $a_c = 0$, $z = 0$, and $t = 0$; $\kappa(\theta, \varepsilon_\kappa) = [\kappa_0 + \kappa_\theta \log\theta + \sigma_\kappa \varepsilon_\kappa]^+ \cdot \bar{V}$ is the psychic cost of college; and σ_{cd} is the logit scale parameter. The integral over $\varepsilon_\kappa \sim N(0,1)$ is computed via Gauss-Hermite quadrature with seven nodes.

Boundary monotonicity correction. At very low parent wealth ($a_p \approx 0$), the coupled HJB can produce spurious college probabilities due to convergence difficulties at the corner of the state space. We enforce monotonicity by scanning from the bottom of the a_p grid upward: if the college probability at point i exceeds that at point $i+1$ for the first few grid points, we cap it at the first interior-monotone value. This correction affects at most the bottom third of the grid.

J.5 Stage 4: Dynastic Simulation

We simulate $M = 4{,}000$ dynasties across $N_{\text{coh}} = 11$ overlapping generations using the Euler-Maruyama method.

Initialization. The first cohort begins at median ability $\theta = \theta_{(N_\theta+1)/2}$. Subsequent generations draw ability from the AR(1) process: $\log \theta' = \rho_\theta \log \theta + \mu_\theta(1-\rho_\theta) + \sigma_\theta \varepsilon$. Each new generation's parent wealth equals the previous generation's terminal child wealth (clamped to $[0, \bar{a}_p]$). The initial income shock z_0 is drawn from the stationary distribution $z_0 \sim \mathcal{N}(0, \sigma_e^2/2\kappa_e)$.

College decision. For each dynasty m in cohort g, we interpolate the college probability $\Pr(\mathrm{C} \mid a_p^{m,g}, \theta^{m,g})$ from the solved grid and draw education status from a Bernoulli distribution.

Forward simulation. Given education choices and initial conditions, we simulate forward using Euler-Maruyama discretization. At each time step $n = 1, \ldots, N_{\text{steps}}$:

1. **Interpolate policies**: trilinear interpolation of (c_p, c_c, τ) from the solved HJB policy grids at the current state (a_p^n, a_c^n, z^n).
2. **Compute income**: parent income y_p (deterministic or Social Security if retired) and child income y_c (with transitory shock $\varepsilon \sim \mathcal{N}(0, \sigma_{\varepsilon,e}^2)$, net of tuition during college).
3. **Update wealth**:

$$\begin{aligned} a_p^{n+1} &= a_p^n + \left(r\, a_p^n + y_p - c_p - \tau\right) \Delta t, \\ a_c^{n+1} &= a_c^n + \left(\tilde{r}_c\, a_c^n + y_c - c_c + \tau\right) \Delta t. \end{aligned} \tag{39}$$

4. **Update income shock**: $z^{n+1} = z^n - \kappa_e\, z^n\, \Delta t + \sigma_e \sqrt{\Delta t}\, \xi^n$, where $\xi^n \sim \mathcal{N}(0,1)$.
5. **Enforce constraints**: clamp a_p to $[0, \bar{a}_p]$ and a_c to $[\underline{a}_c, \bar{a}_c]$ (with $\underline{a}_c = -\bar{b}$ during college, $\underline{a}_c = 0$ after). Clamp z to the grid bounds.

Reproducibility. All random draws (initial z_0, college choice, OU innovations, transitory shocks) are pre-generated sequentially before the threaded simulation loop, ensuring bit-identical results regardless of the number of parallel threads.

J.6 Stage 5: Moment Computation and SMM Estimation

The model is estimated by Simulated Method of Moments using 35 target moments organized into five blocks. All moments are computed from a settled cohort (the second-to-last generation) to avoid transient dynamics from initial conditions.

Moment blocks.

1. **College attendance** (16 moments): enrollment rates by parent-wealth quartile $\times$ ability quartile.
2. **Income** (3 moments): HS/college income ratio, cross-sectional income standard deviation for HS and college workers at age 35.
3. **Transfers** (5 moments): mean cumulative transfer during college years, transfer probability $\Pr(\tau > 0)$ by parent-wealth quartile.
4. **Wealth** (10 moments): wealth-to-income ratio, intergenerational rank-rank slope, conditional child wealth $\mathbb{E}[a_c \mid \text{parent wealth Q}]$ (4 moments), median wealth for parents and children, variance of $\text{IHS}(a)$ for parents and children.
5. **Decumulation** (1 moment): parent wealth decumulation rate from age 60 to 70.

Loss function. The objective uses block-diagonal weighting so that each group contributes proportionally:

$$\mathcal{L}(\boldsymbol{\vartheta}) = \sum_{g=1}^{5} \frac{w_g}{n_g} \sum_{m \in g} \left(\frac{m_m(\boldsymbol{\vartheta}) - \hat{m}_m}{\hat{m}_m} \right)^2, \tag{40}$$

where n_g is the number of moments in block g and w_g is the group weight (college moments receive $w_g = 3$; all other groups receive $w_g = 1$).

Optimization. We minimize $\mathcal{L}$ over 17 parameters using a custom batch evolutionary optimizer with distributed workers. Each worker evaluates one parameter vector by solving the full model (Stages 1–4) and computing the loss. Multi-start initialization ensures coverage of the parameter space.

J.7 Stage 6: Counterfactual Economies

Full commitment. The parent and child commit to a transfer schedule $\{\tau_t\}_{t\geq 0}$ at $t = 0$. This is solved as a single planner's problem that jointly maximizes parent and child welfare, eliminating the Samaritan's dilemma. The resulting college rates isolate the moral hazard effect: $\Delta_{\text{MH}} = \Pr(\text{C} \mid \text{baseline}) - \Pr(\text{C} \mid \text{FC})$.

No-moral-hazard (unconditional). The parent makes a one-time lump-sum transfer τ_0 at $t = 0$, after which the child solves the child-alone problem for their entire remaining life. The child receives τ_0 regardless of education choice. The parent's problem is:

$$\max_{\tau_0 \in [0, a_p]} V_p^{\text{alone}}(a_p - \tau_0) + \eta \cdot \mathbb{E}_{\varepsilon_\kappa}\left[\max\left\{V^{c,\text{alone}}(\tau_0 \mid \text{C}) - \kappa(\theta, \varepsilon_\kappa) + \varepsilon_C,\ V^{c,\text{alone}}(\tau_0 \mid \text{HS}) + \varepsilon_{HS}\right\}\right].$$

No-moral-hazard (menu). Same as above, but the parent commits to an education-contingent menu: a transfer τ_C paid if the child attends college and τ_{HS} paid if the child chooses high school, both optimized in $[0, a_p]$. This is the efficient one-time-transfer benchmark; it nests the unconditional transfer ($\tau_C = \tau_{HS}$) and measures the effect of removing moral hazard *plus* optimal education targeting.

Four-way decomposition. The difference in college rates across these allocations decomposes as:

$$\underbrace{\Pr(\text{C} \mid \text{BL}) - \Pr(\text{C} \mid \text{menu})}_{\Delta_{\text{MH}}^{\text{total}}} = \underbrace{\Pr(\text{C} \mid \text{BL}) - \Pr(\text{C} \mid \text{uncond.})}_{\Delta_{\text{MH}}^{\text{pure}}} + \underbrace{\Pr(\text{C} \mid \text{uncond.}) - \Pr(\text{C} \mid \text{menu})}_{\Delta_{\text{targ}}}. \tag{41}$$

The signs are consistent with the main text (Section 7): a positive $\Delta_{\text{MH}}^{\text{pure}}$ indicates that baseline moral hazard raises college attendance relative to the no-moral-hazard economy.